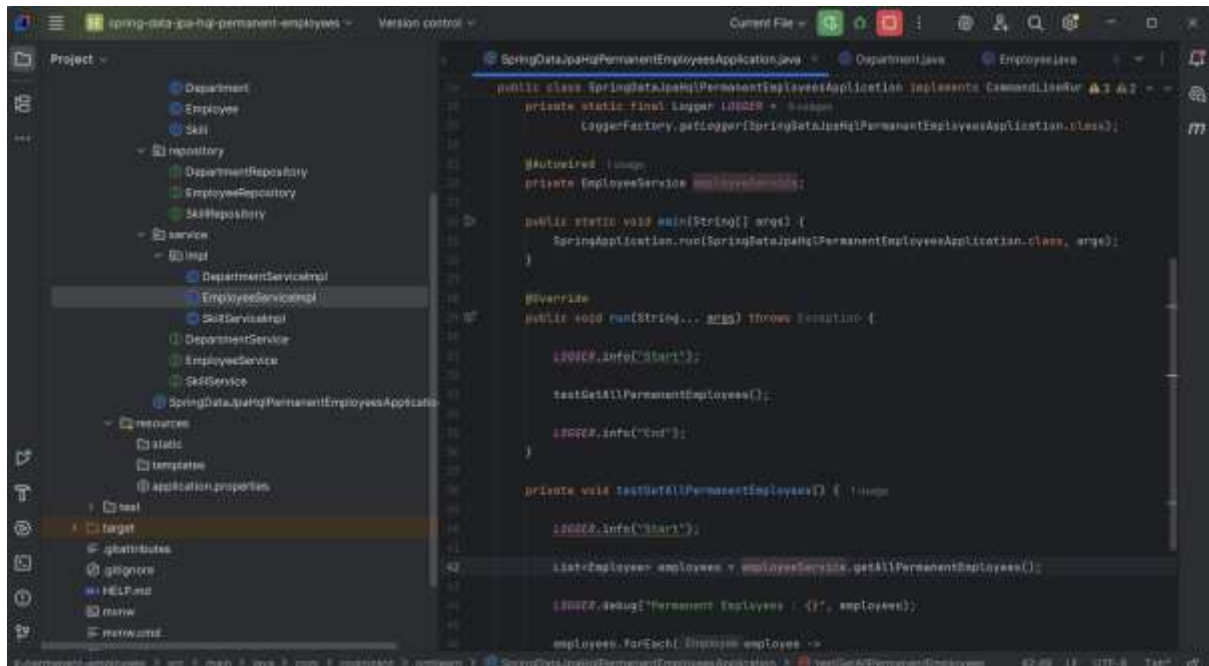


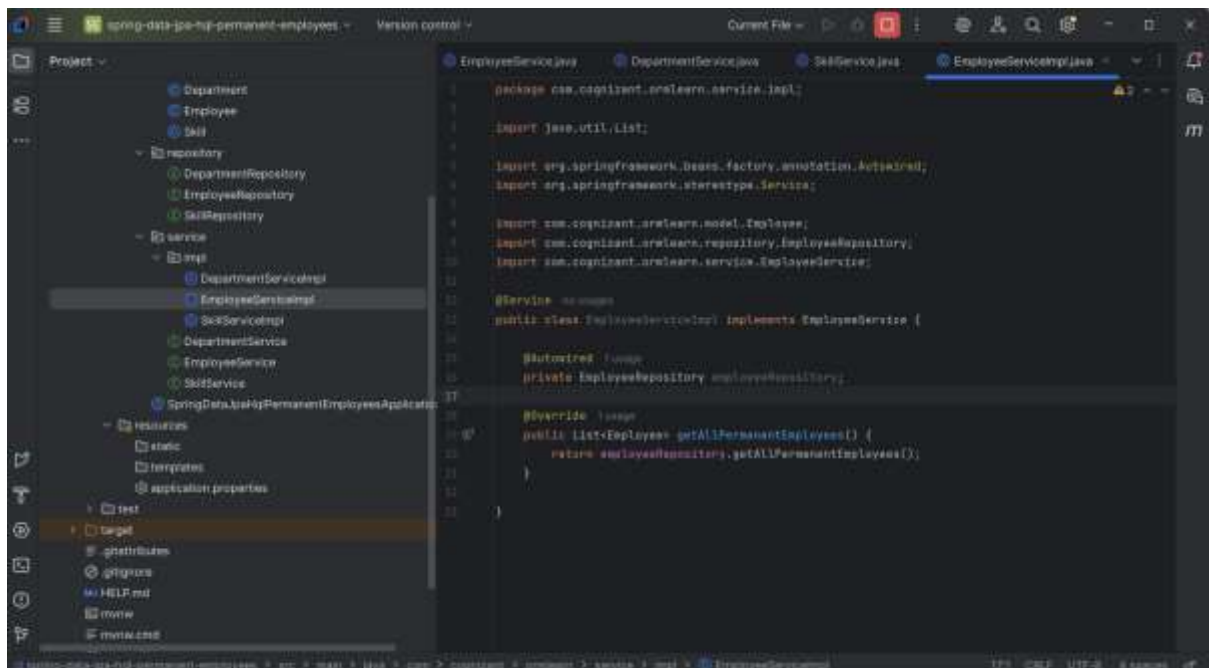
Hands on 2

Get all permanent employees using HQL

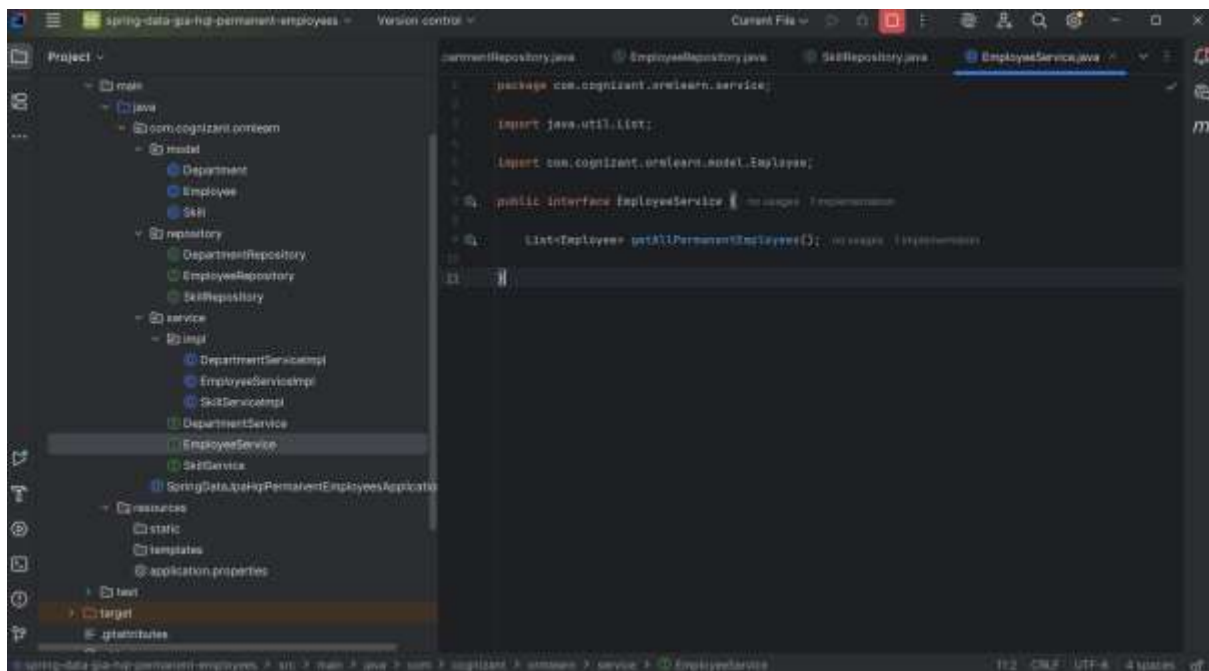
Step 1: Basic HQL query



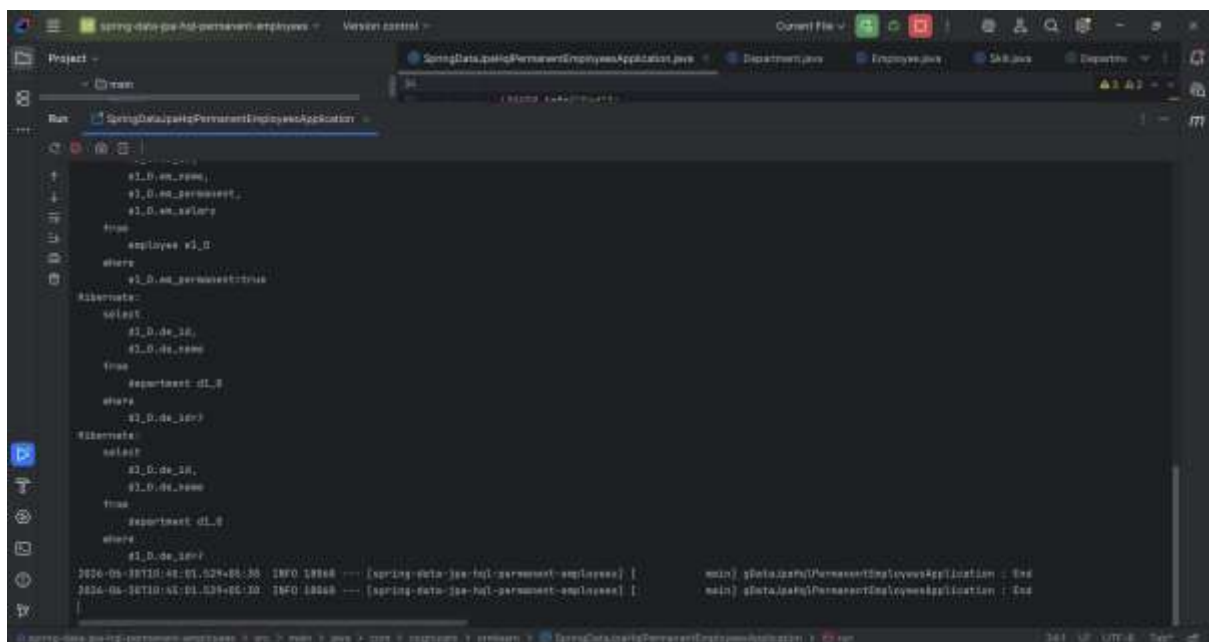
```
1  package com.cognizant.armlern.service.impl;
2
3  import java.util.List;
4
5  import org.springframework.beans.factory.annotation.Autowired;
6  import org.springframework.stereotype.Service;
7
8  import com.cognizant.armlern.model.Employee;
9  import com.cognizant.armlern.repository.EmployeeRepository;
10 import com.cognizant.armlern.service.EmployeeService;
11
12 @Service
13 public class EmployeeServiceImpl implements EmployeeService {
14
15     @Autowired
16     private EmployeeRepository employeeRepository;
17
18     @Override
19     public List<Employee> getAllPermanentEmployees() {
20         return employeeRepository.getAllPermanentEmployees();
21     }
22 }
```



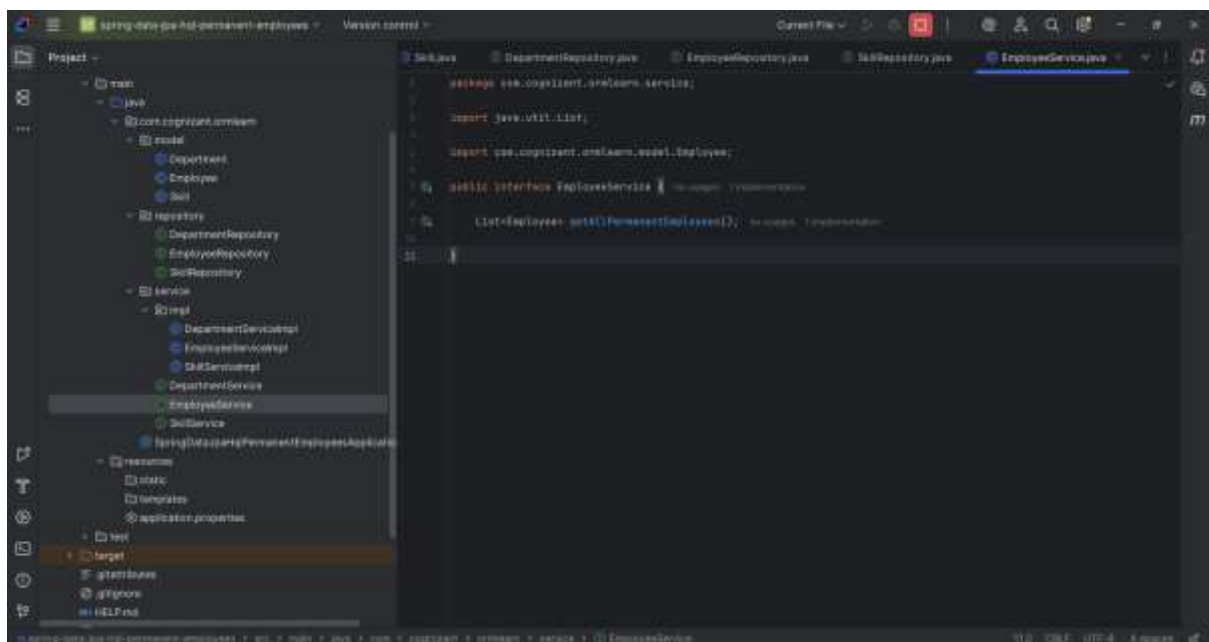
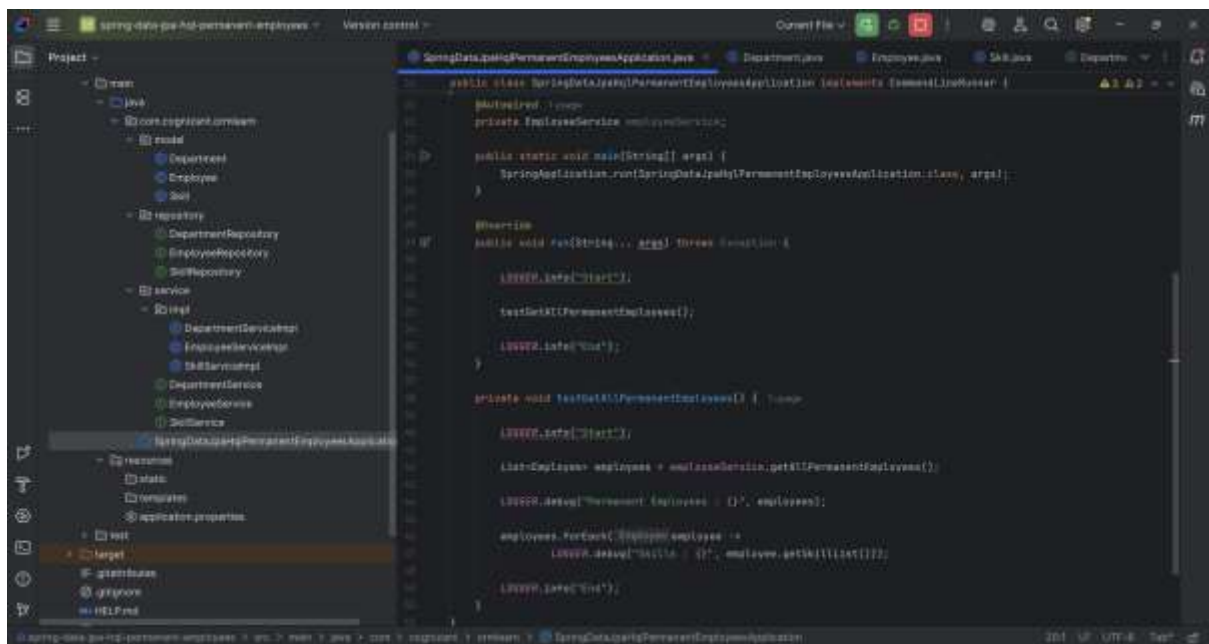
```
1  package com.cognizant.armlern.service.impl;
2
3  import java.util.List;
4
5  import org.springframework.beans.factory.annotation.Autowired;
6  import org.springframework.stereotype.Service;
7
8  import com.cognizant.armlern.model.Employee;
9  import com.cognizant.armlern.repository.EmployeeRepository;
10 import com.cognizant.armlern.service.EmployeeService;
11
12 @Service
13 public class EmployeeServiceImpl implements EmployeeService {
14
15     @Autowired
16     private EmployeeRepository employeeRepository;
17
18     @Override
19     public List<Employee> getAllPermanentEmployees() {
20         return employeeRepository.getAllPermanentEmployees();
21     }
22 }
```

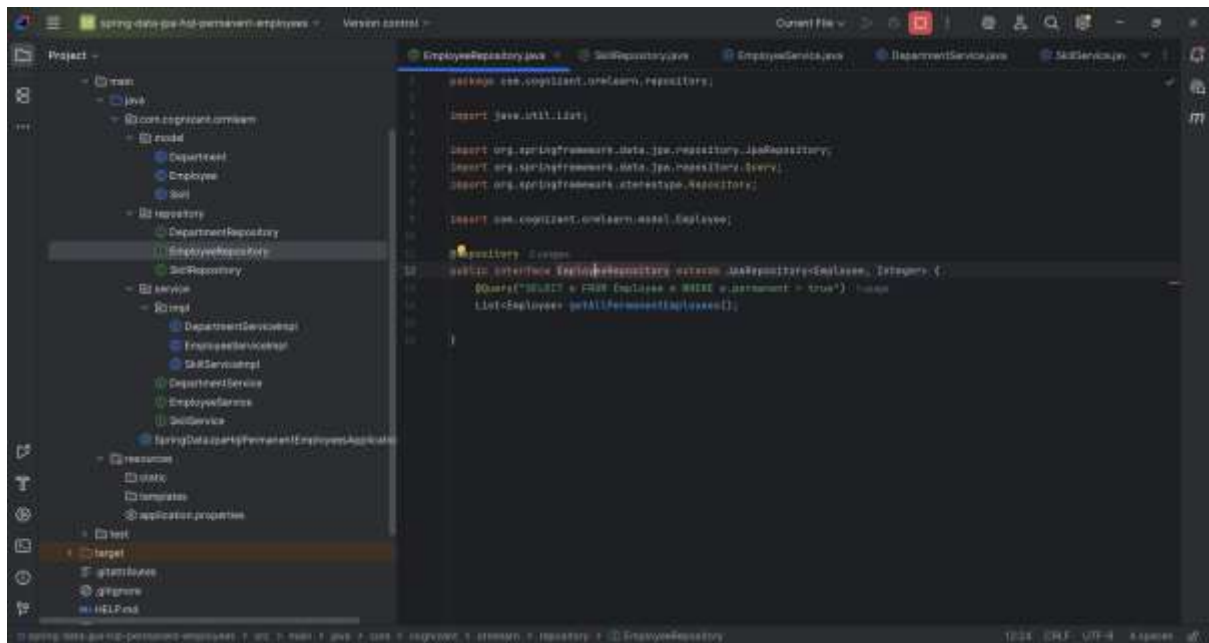
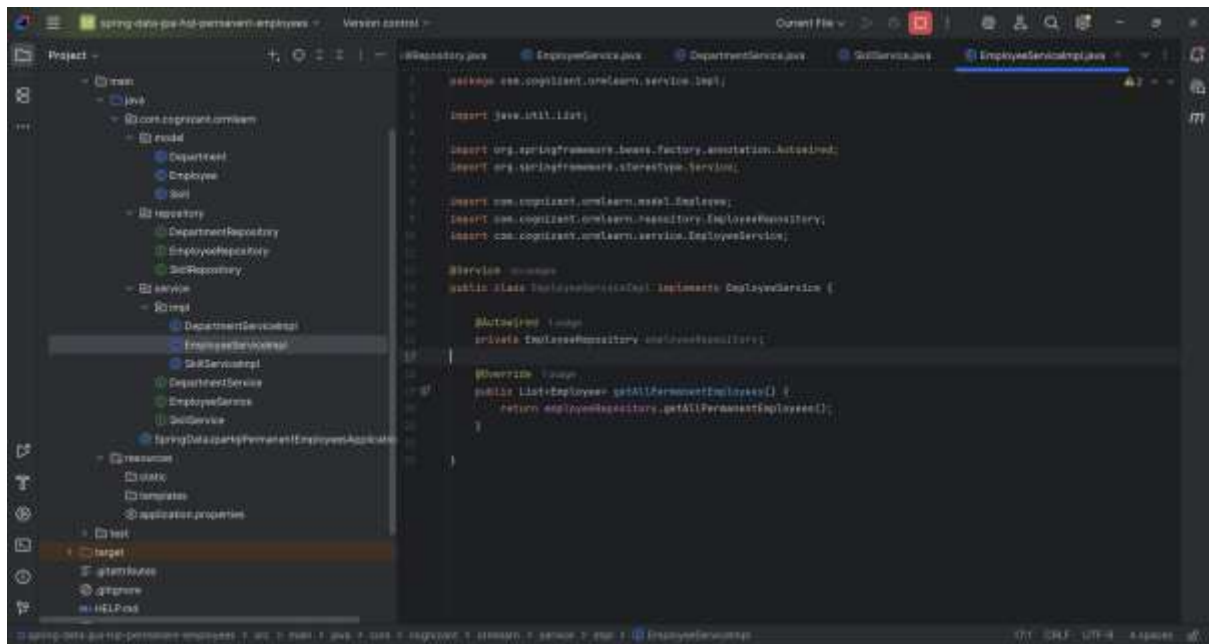



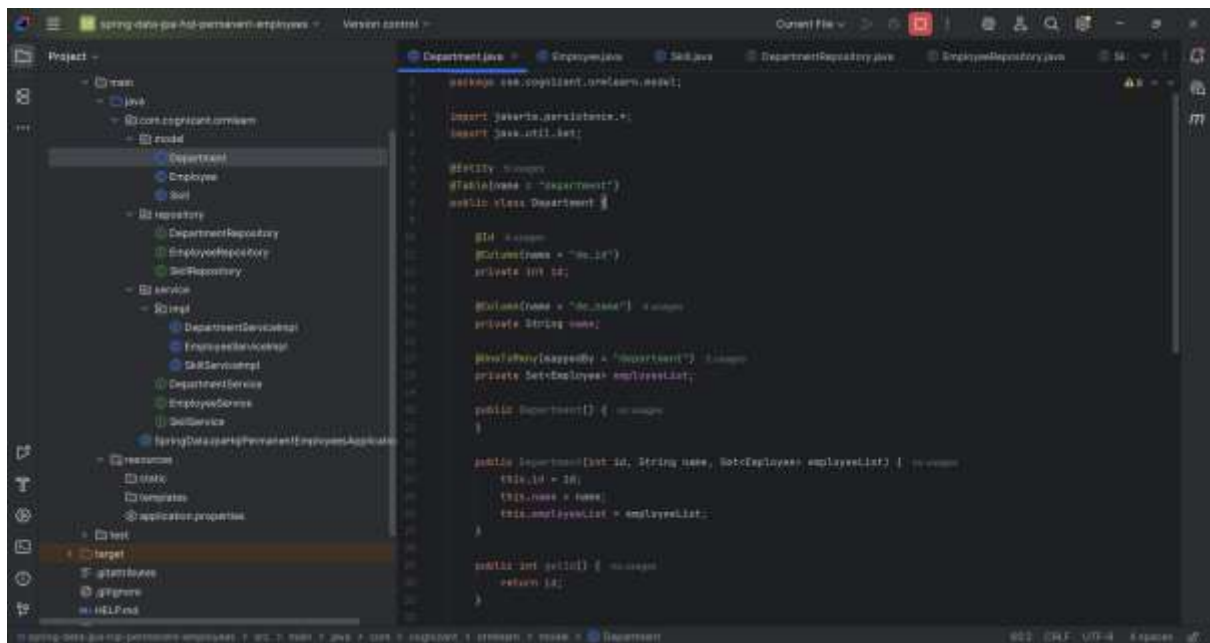
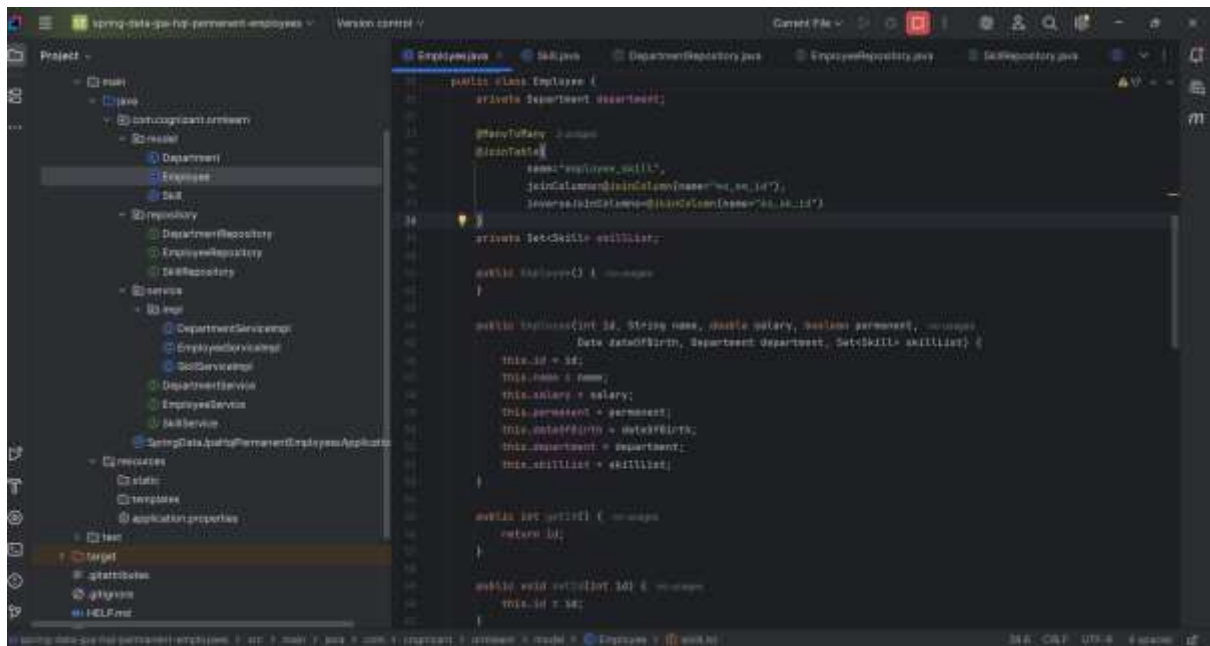
OUTPUT:

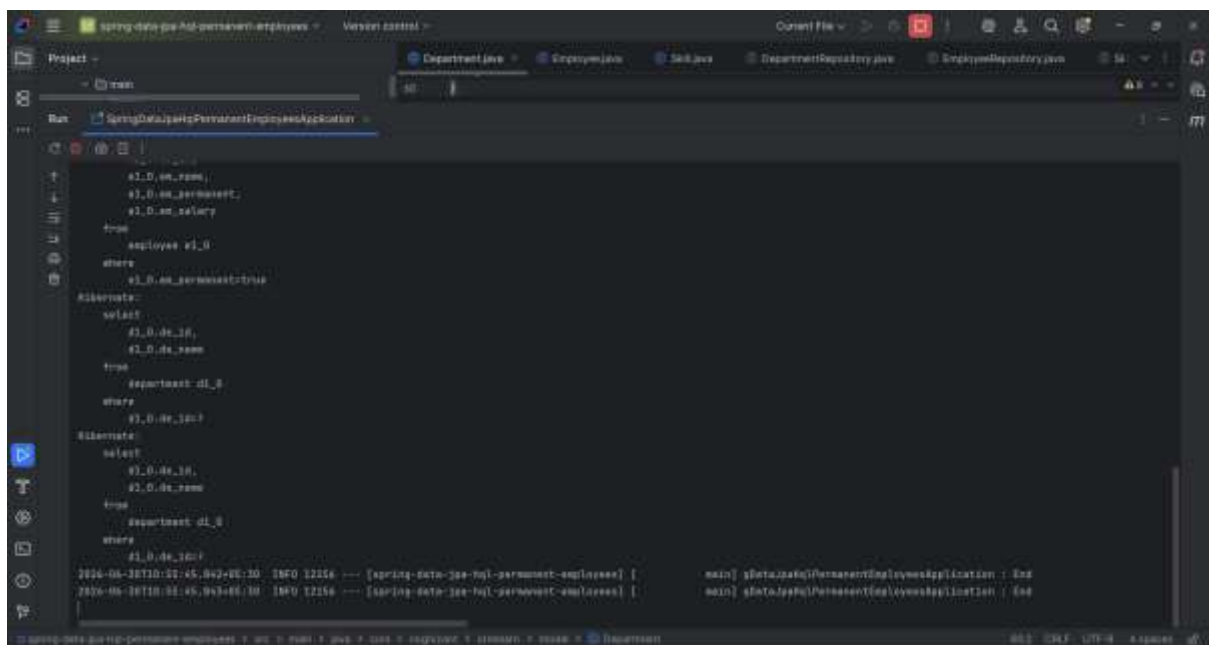
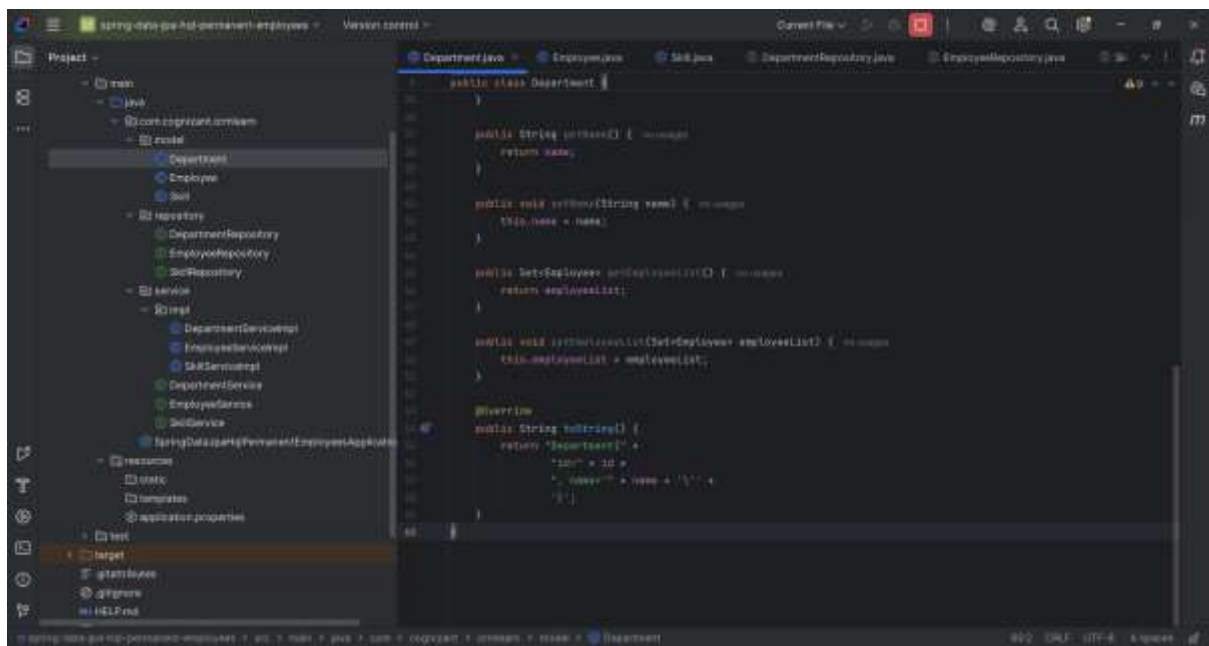


Step 2: Remove EAGER fetch configuration

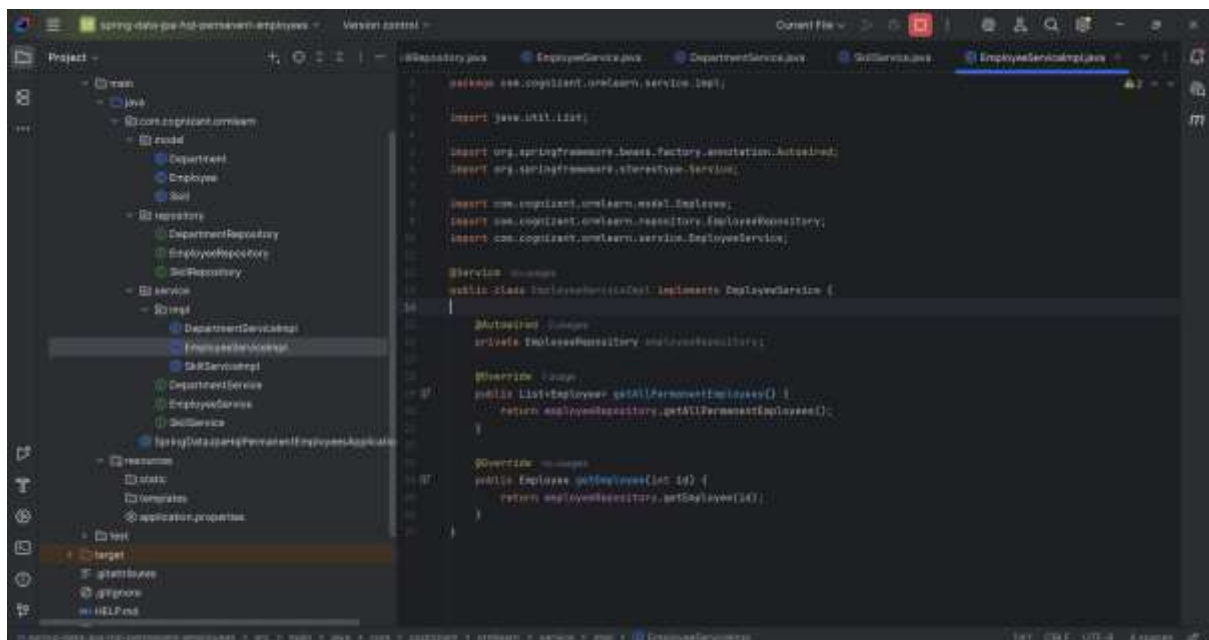
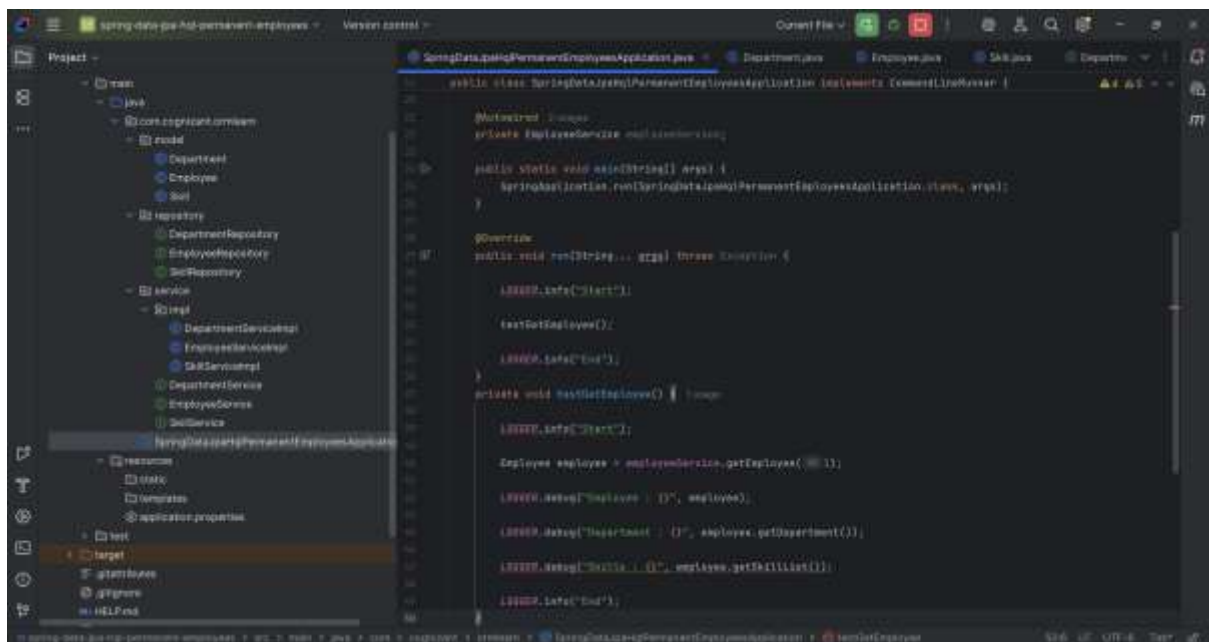


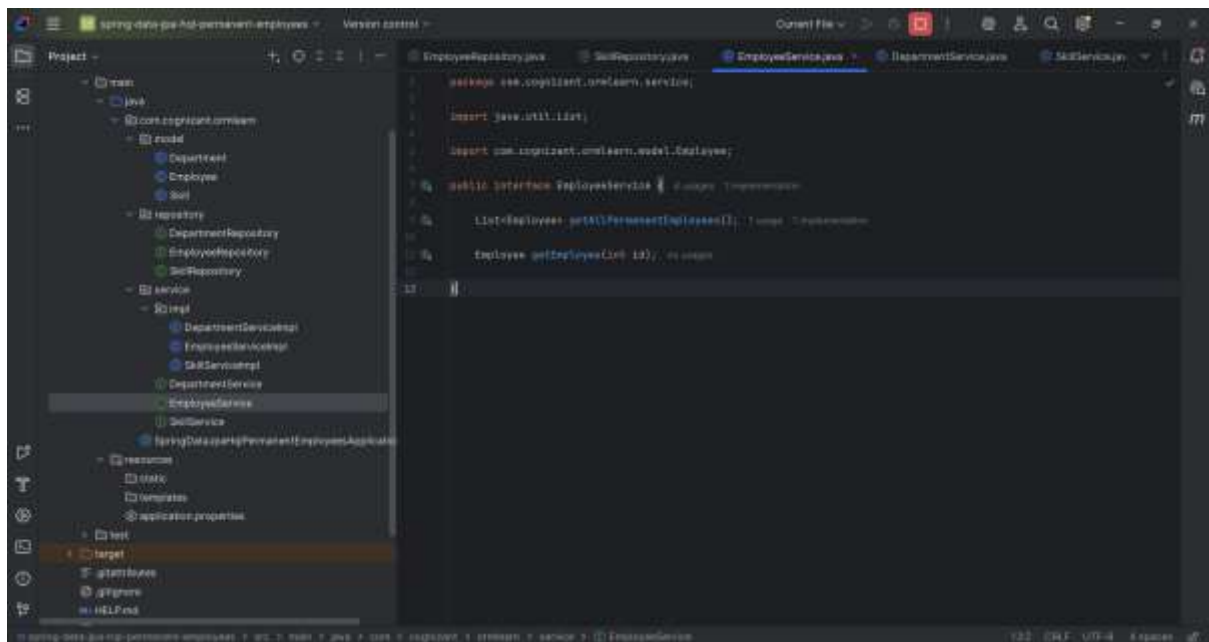
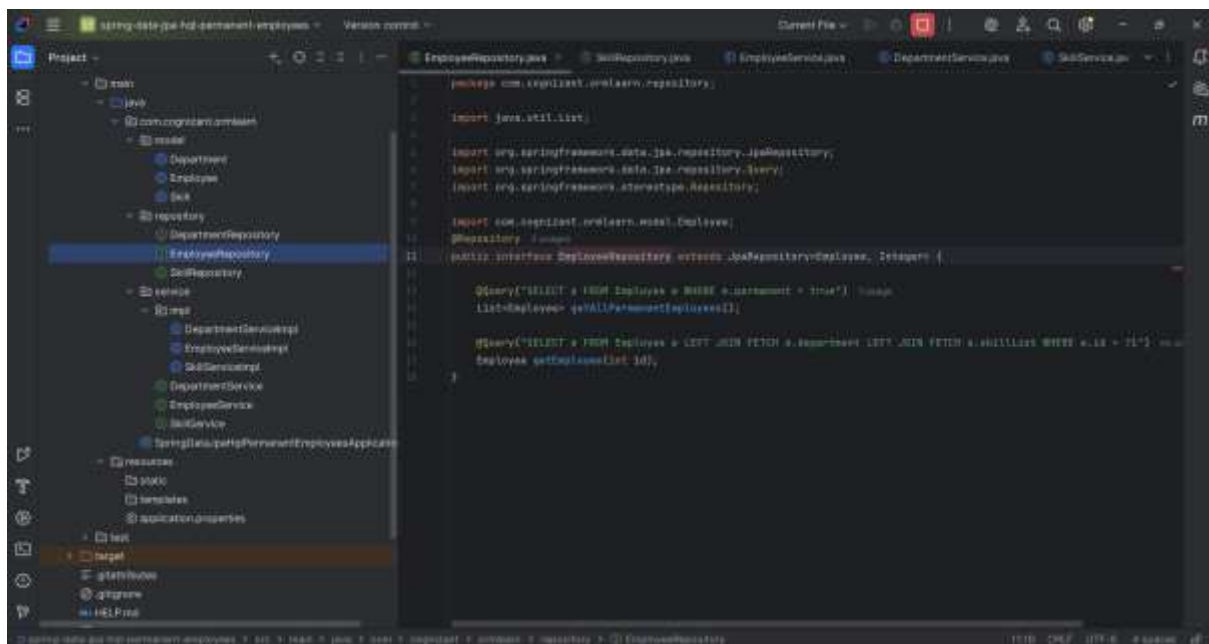


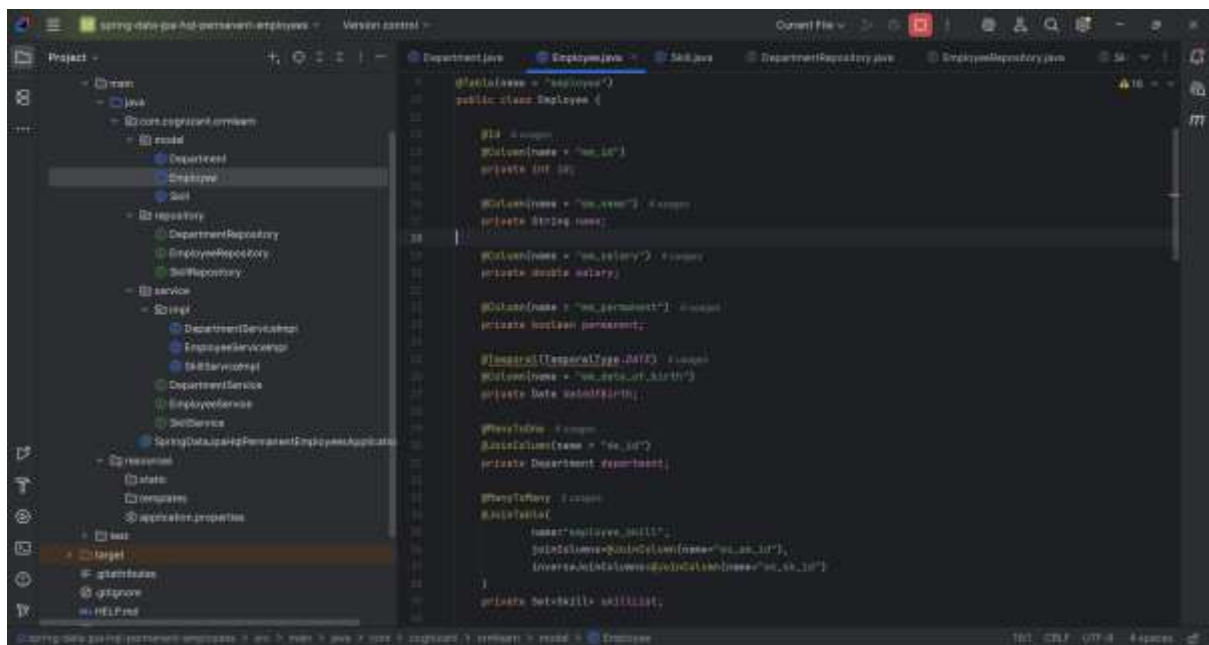
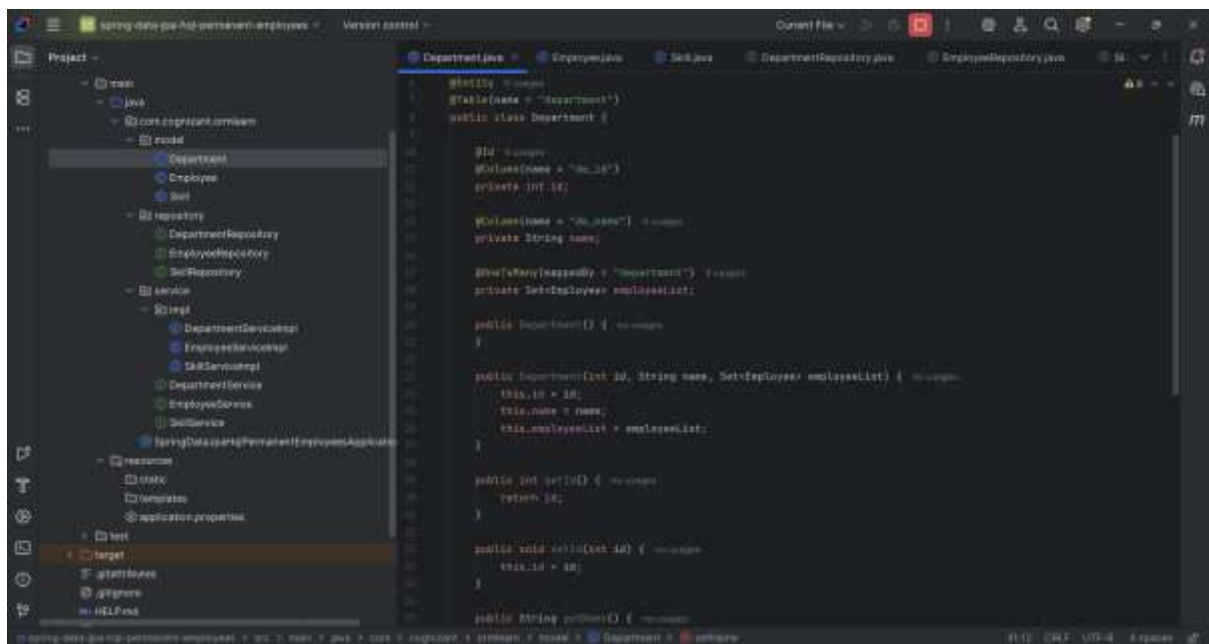


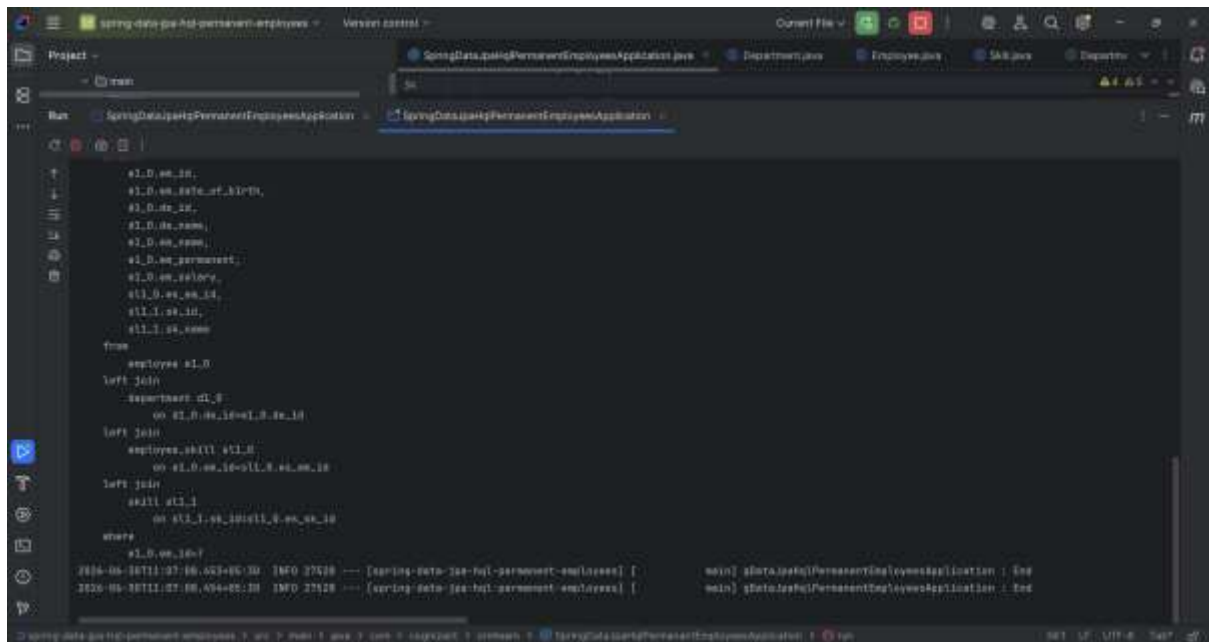


Step 3: Optimize using LEFT JOIN FETCH



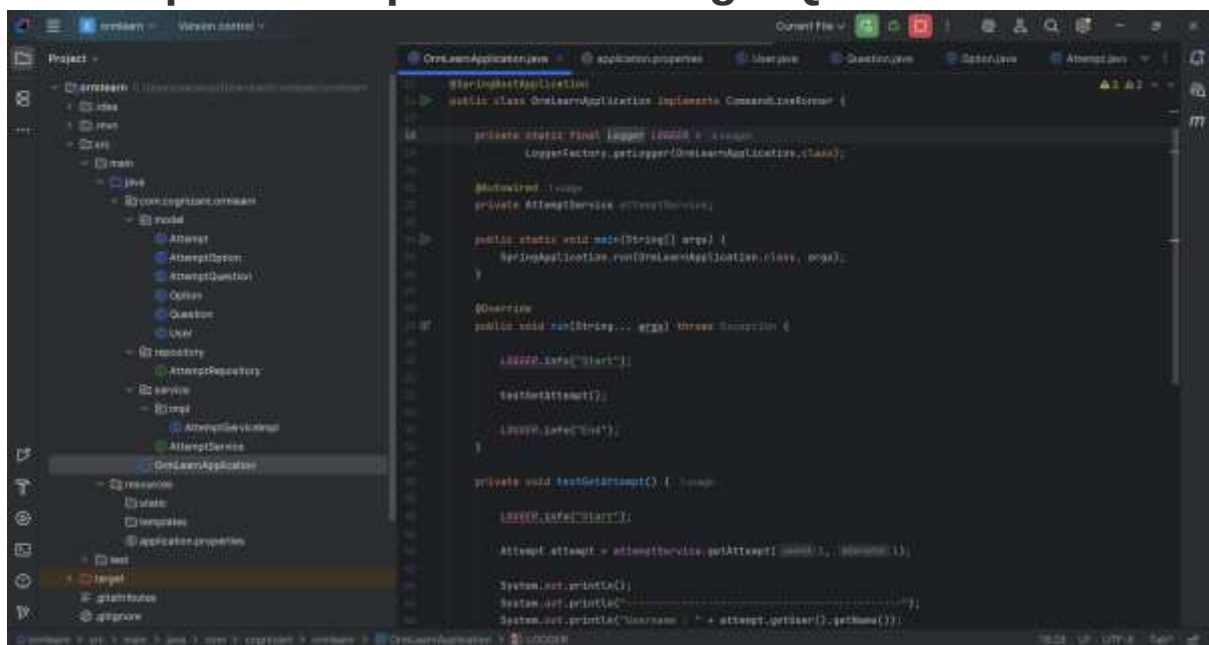


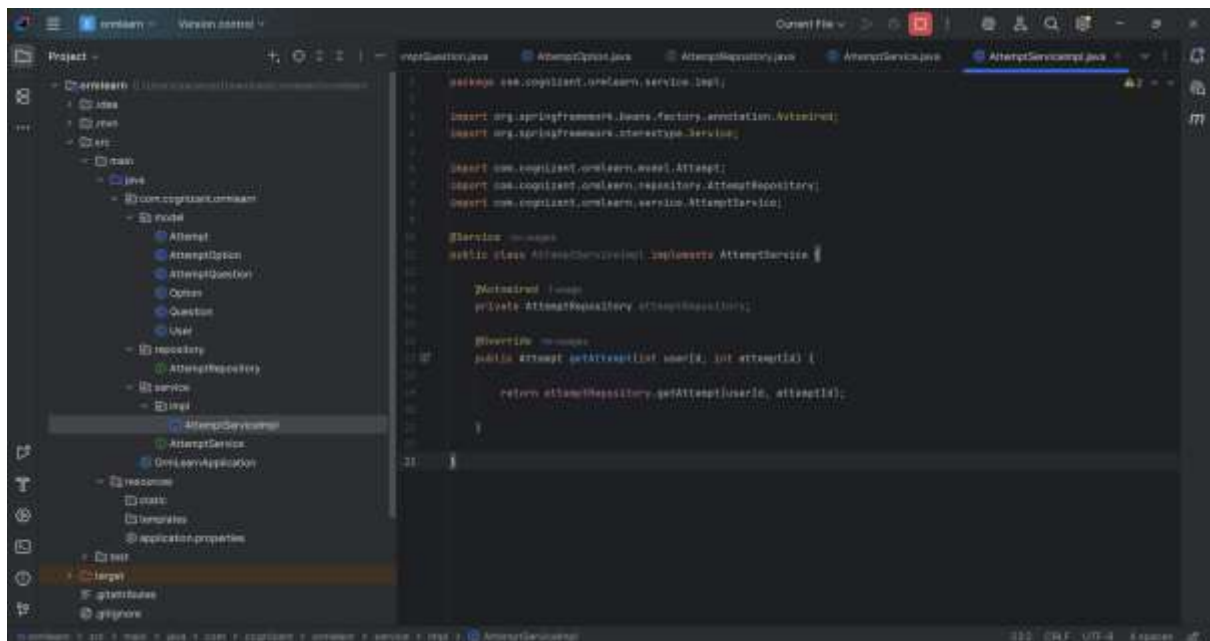
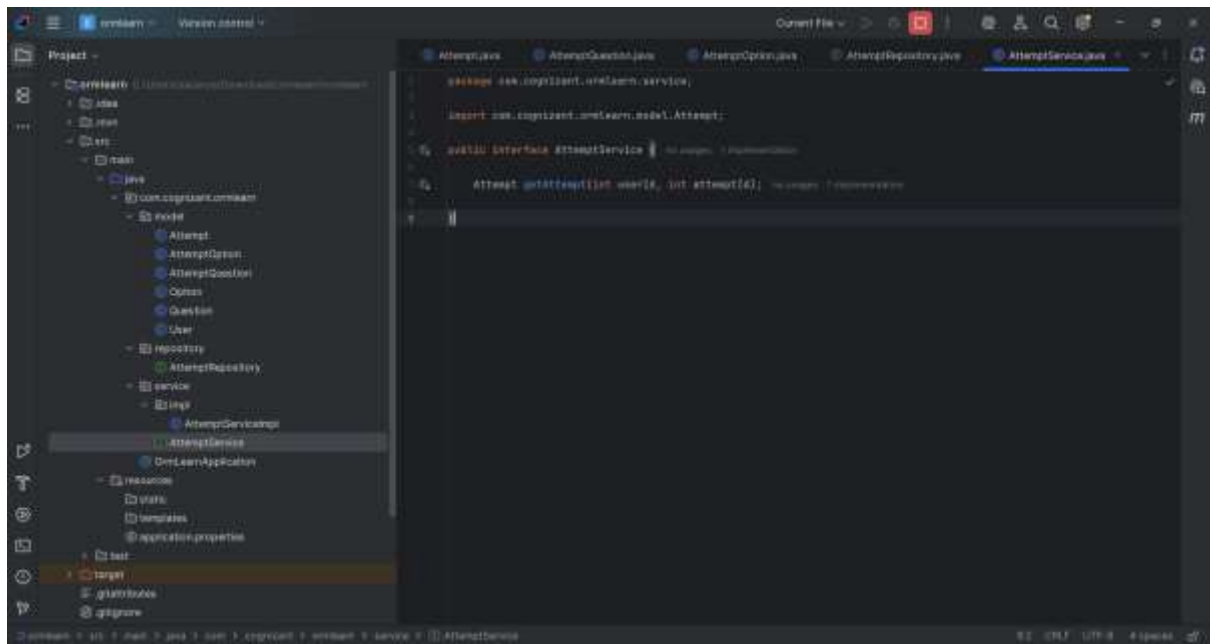




Hands on 3

Fetch quiz attempt details using HQL





```

package com.cognizant.onlearn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.query.Param;

import com.cognizant.onlearn.model.Attempt;

public interface AttemptRepository extends JpaRepository<Attempt, Integer> {

    @Query("SELECT DISTINCT a FROM Attempt a WHERE a.userId = :userId")
    List<Attempt> findByUserId(@Param("userId") Integer userId);

    Attempt findById(@Param("id") Integer id);
}

```

```

@Entity
@Table(name = "user")
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Integer id;

    @Column(name = "name")
    private String name;

    @Column(name = "email")
    private String email;

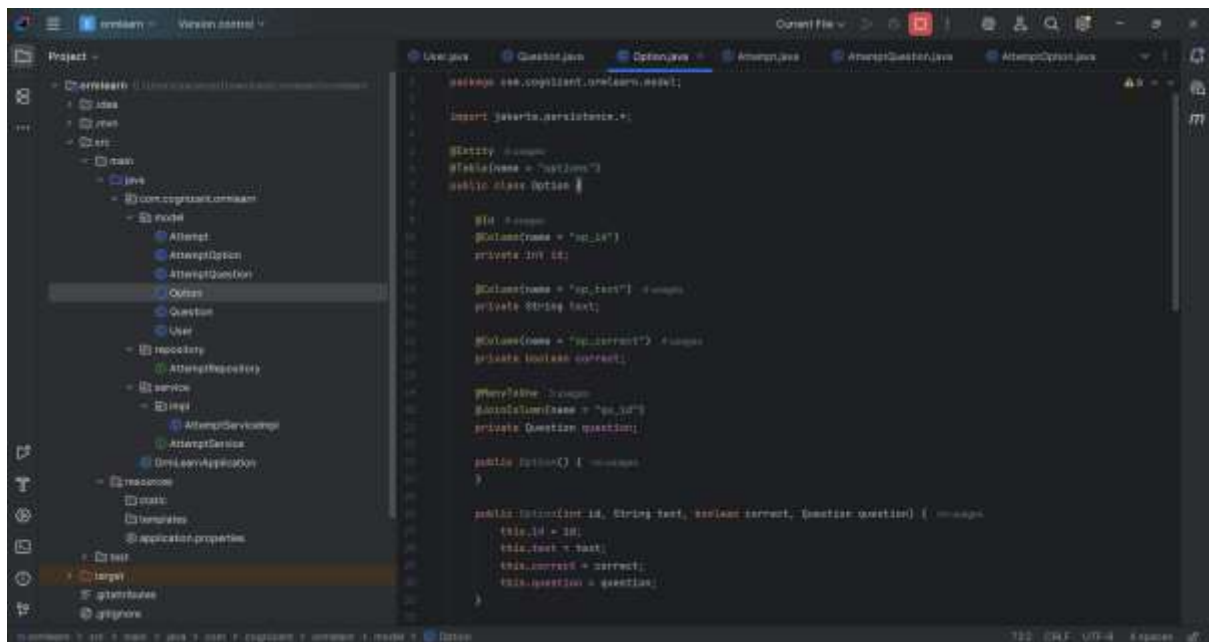
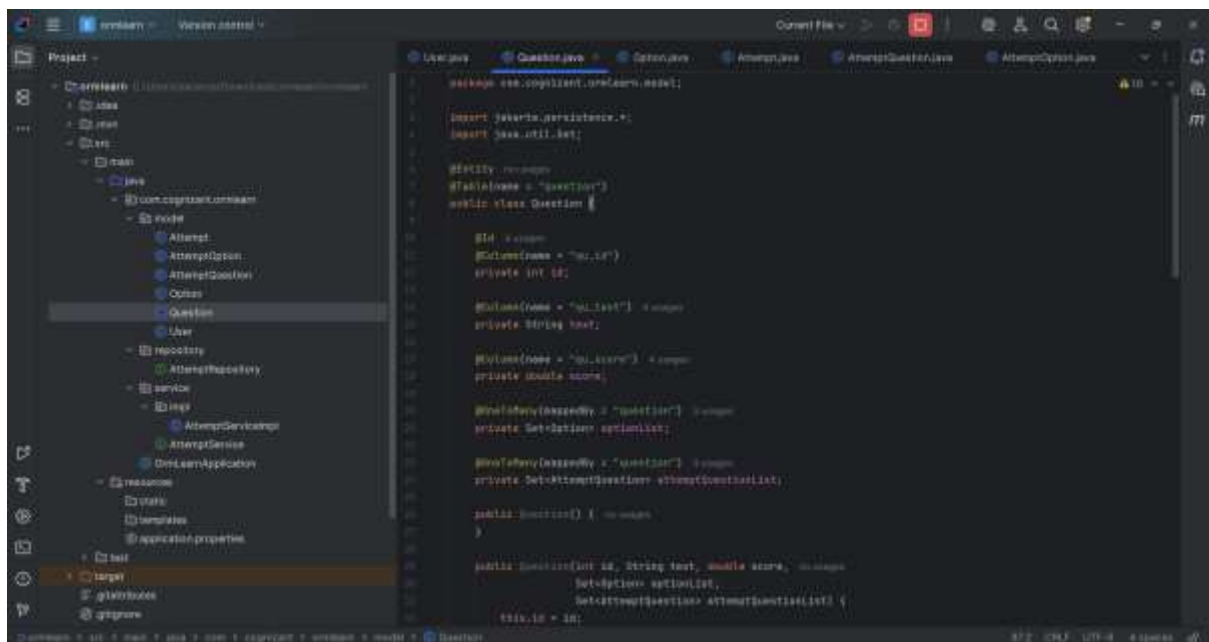
    @OneToMany(mappedBy = "user", cascade = CascadeType.ALL)
    private Set<Attempt> attempts;

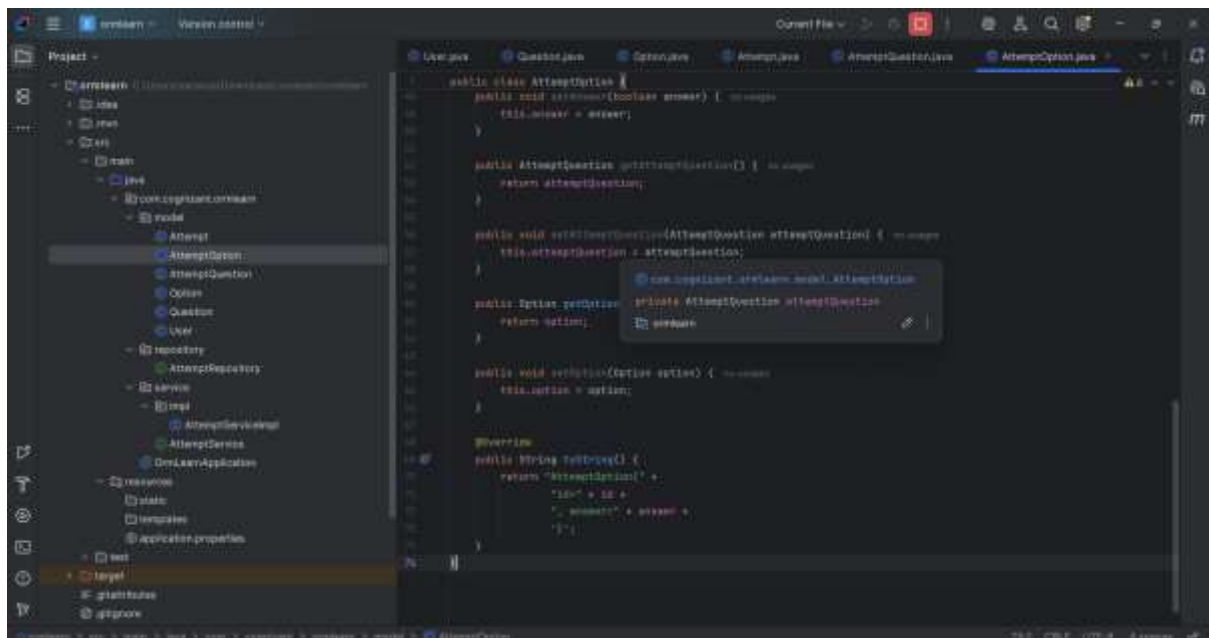
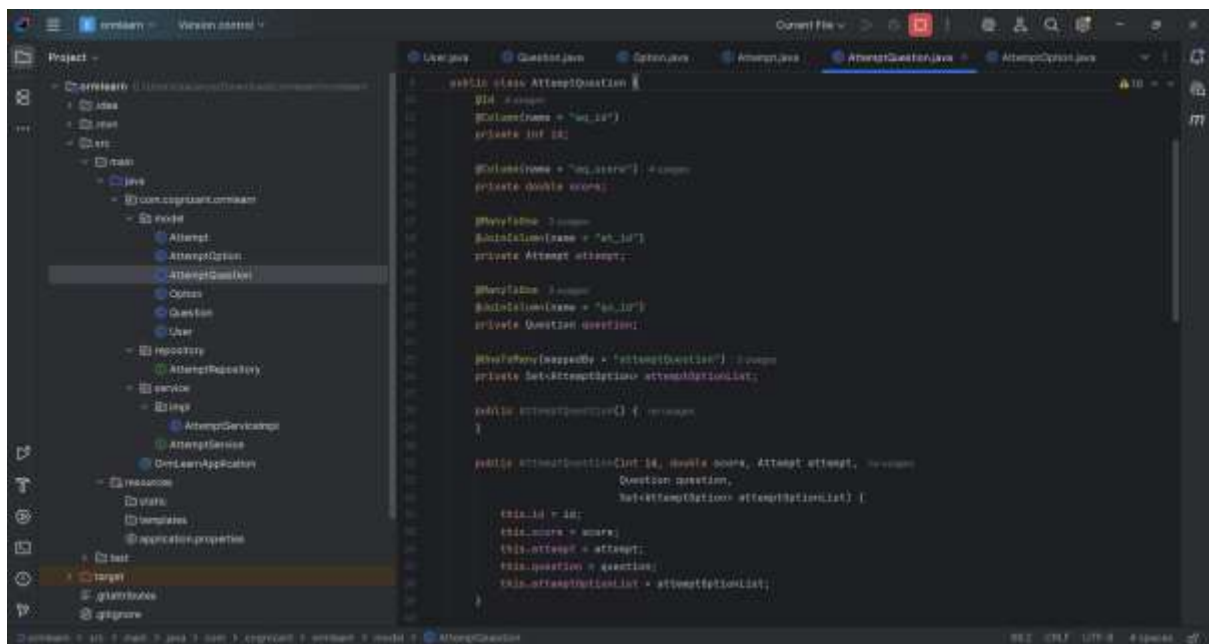
    public User() {}

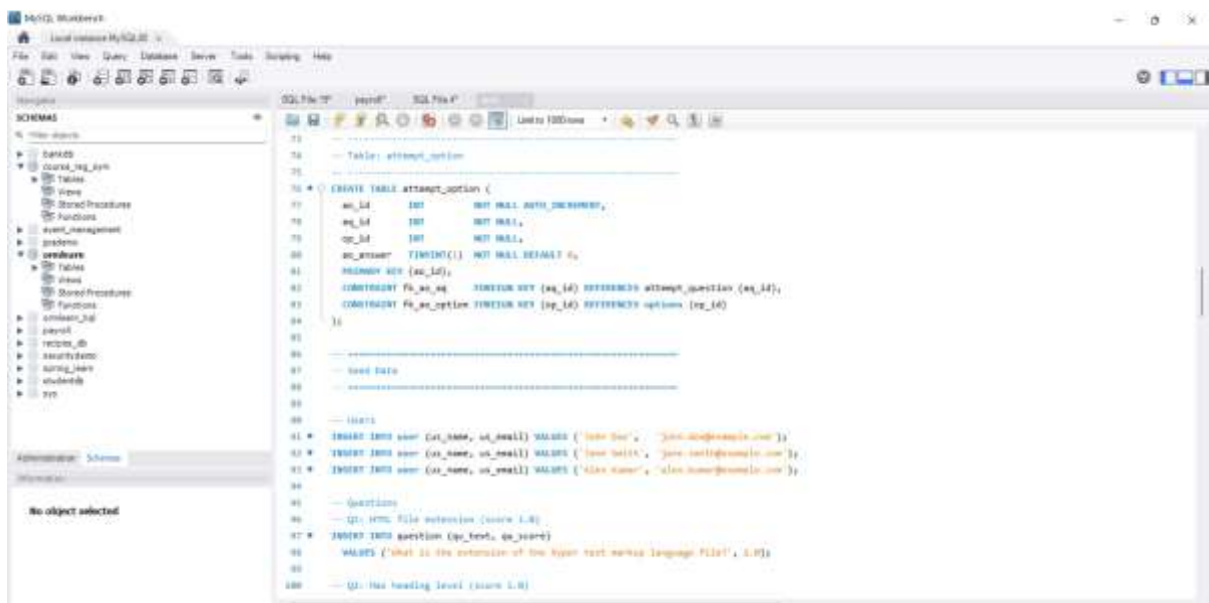
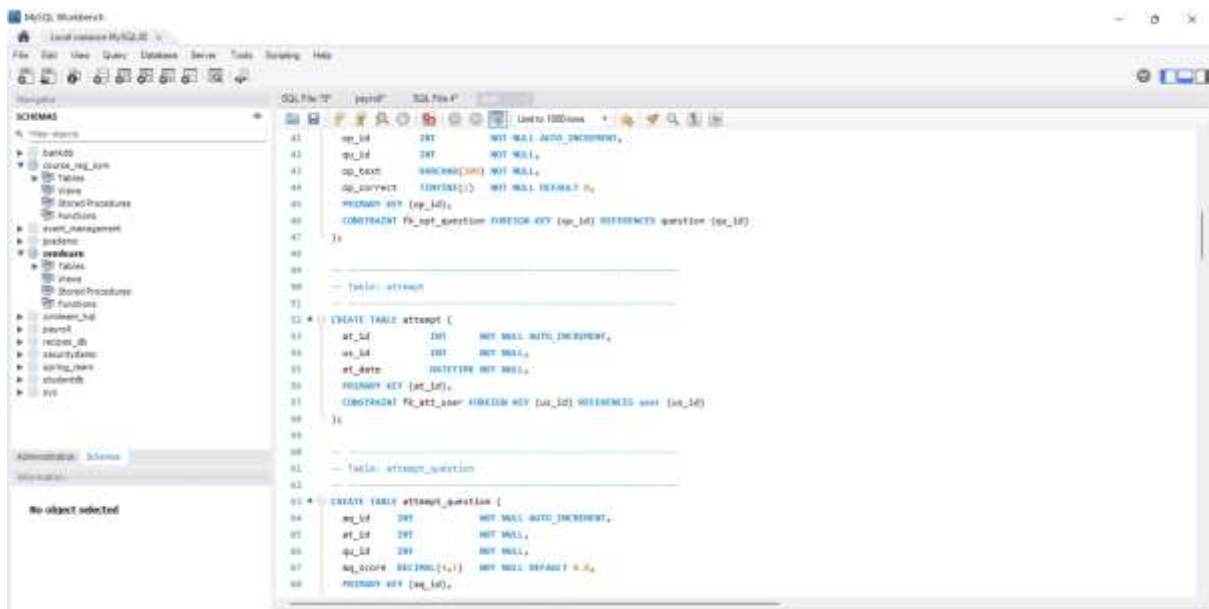
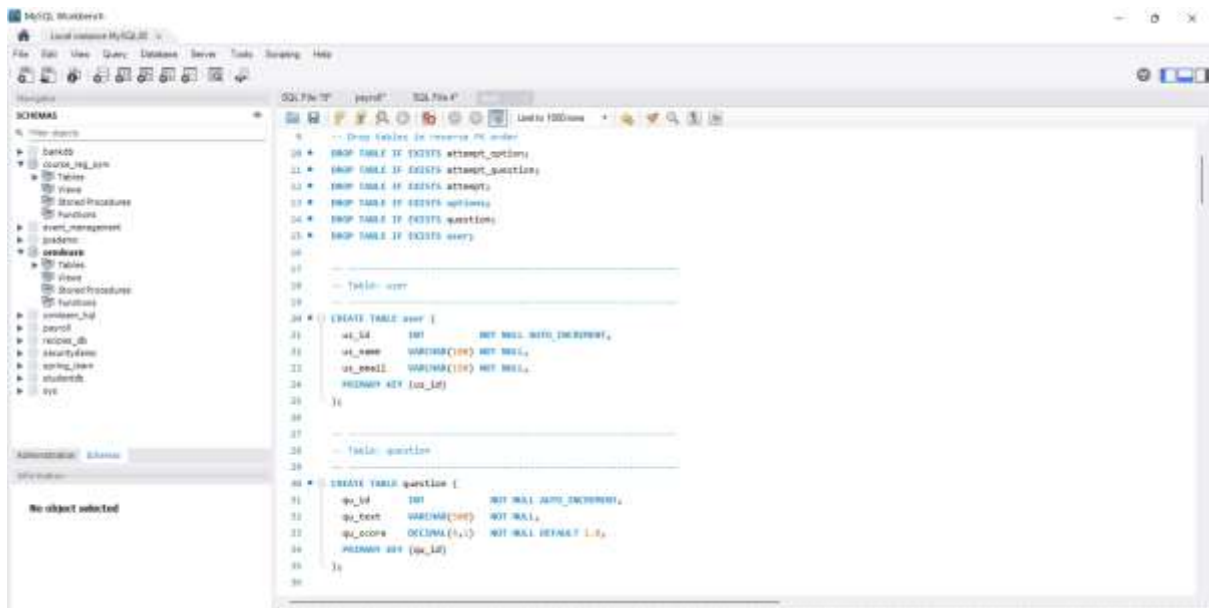
    public User(Integer id, String name, String email, Set<Attempt> attempts) {
        this.id = id;
        this.name = name;
        this.email = email;
        this.attempts = attempts;
    }

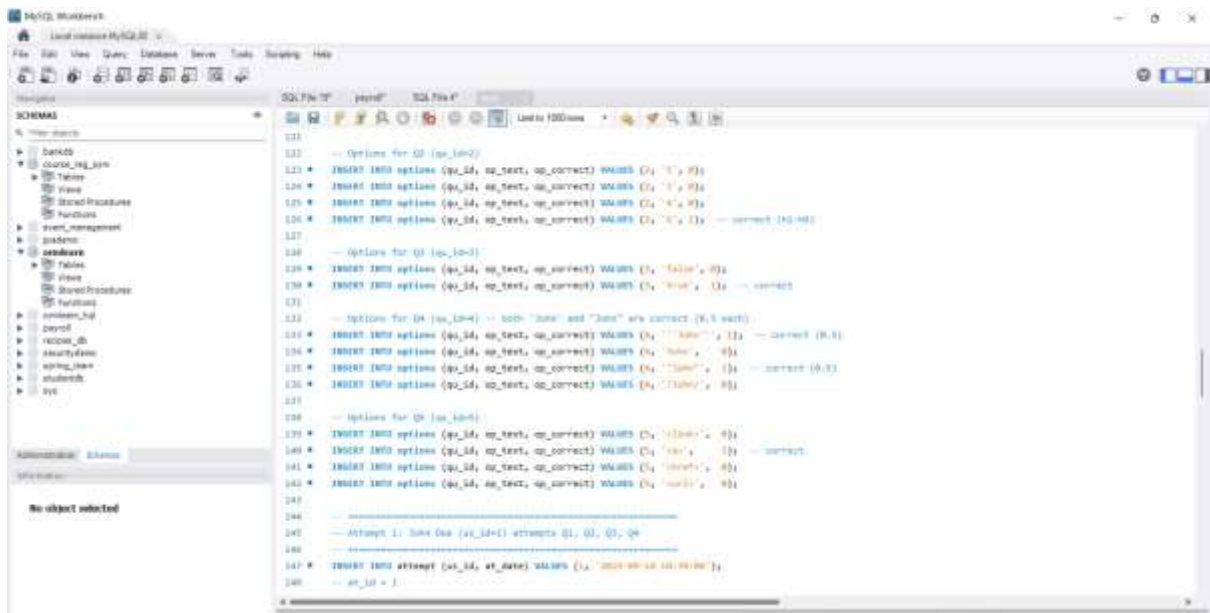
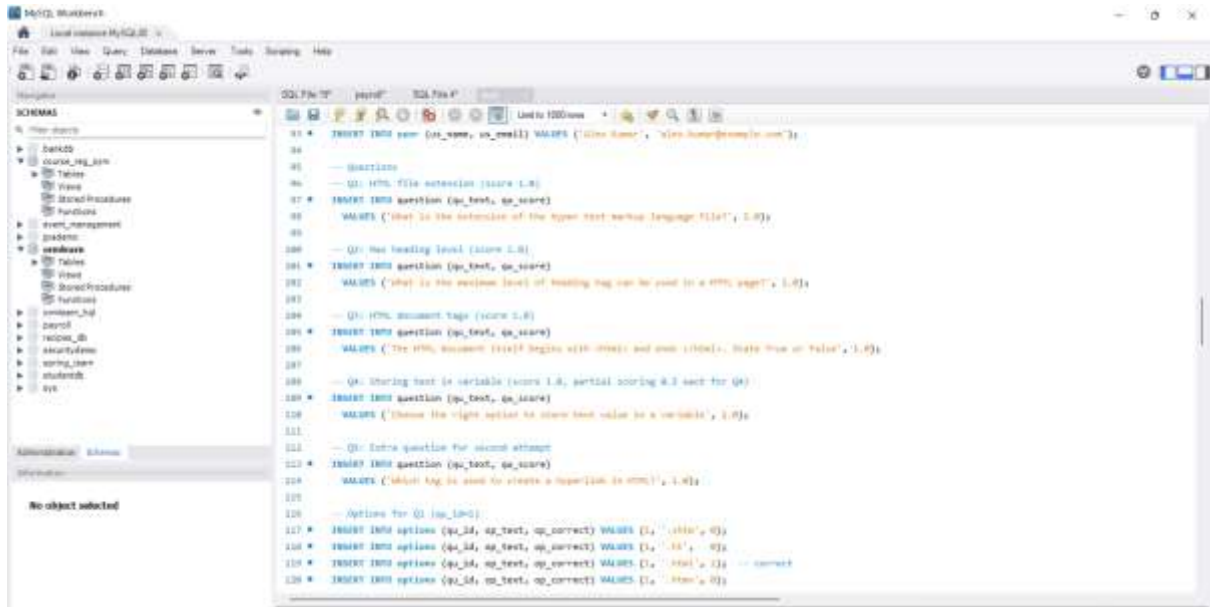
    public Integer getId() {
        return id;
    }
}

```









MySQL Workbench

Local instance MySQL8.0

File Edit View Query Database Server Tools Window Help

SQL Fiddle MySQL 8.0.16

Schema: No object selected

```

149 -- attempt_question row for attempt 1
150 -- attempt_question row for attempt 1
151 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 1, 1.0); -- sq_id=1 Q1 correct
152 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 2, 4.0); -- sq_id=2 Q2 wrong
153 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 3, 1.0); -- sq_id=3 Q3 correct
154 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 4, 0.5); -- sq_id=4 Q4 partial
155
156 -- attempt_option row for attempt 1, Q1 (sq_id=1)
157 -- sq_id 1=0, 2=0, 3=0, 4=0, 5=0, 6=0, 7=0, 8=0, 9=0, 10=0
158 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 1, 0); -- 0 not selected
159 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 2, 0); -- 0 not selected
160 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 3, 1); -- 1 total selected (correct)
161 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 4, 0); -- 0 not selected
162
163 -- attempt_option row for attempt 1, Q2 (sq_id=2)
164 -- sq_id 1=0, 2=0, 3=0, 4=0, 5=0, 6=0, 7=0, 8=0, 9=0, 10=0
165 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 1, 0); -- 0 not selected
166 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 2, 1); -- 1 selected (wrong)
167 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 3, 0); -- 0 not selected
168 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 4, 0); -- 0 not selected
169
170 -- attempt_option row for attempt 1, Q3 (sq_id=3)
171 -- sq_id 1=0, 2=0, 3=0, 4=0, 5=0, 6=0, 7=0, 8=0, 9=0, 10=0
172 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 1, 0); -- 0 not selected
173 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 2, 1); -- 1 total selected (correct)
174
175 -- attempt_option row for attempt 1, Q4 (sq_id=4)
176 -- sq_id 1=0, 2=0, 3=0, 4=0, 5=0, 6=0, 7=0, 8=0, 9=0, 10=0

```

MySQL Workbench

Local instance MySQL8.0

File Edit View Query Database Server Tools Window Help

SQL Fiddle MySQL 8.0.16

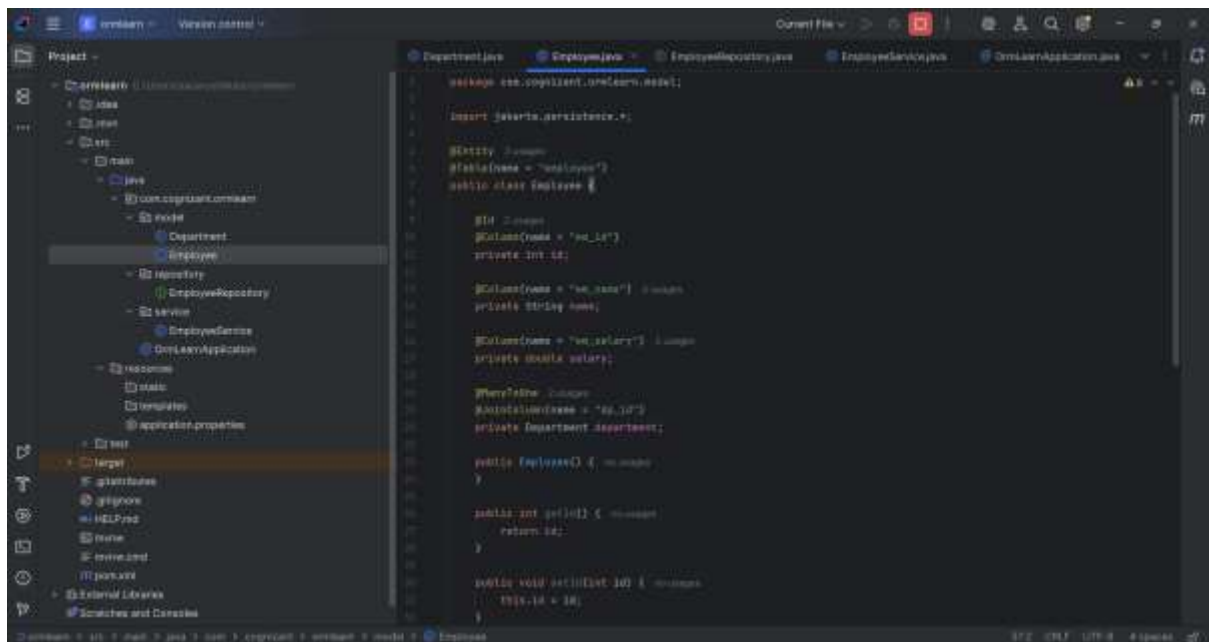
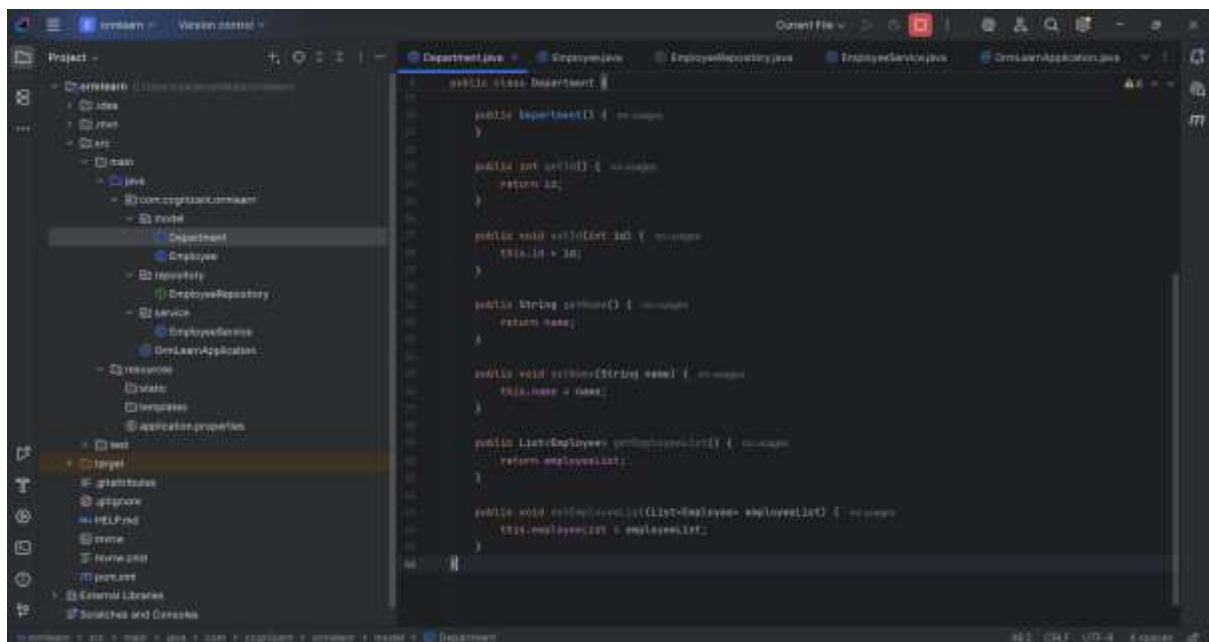
Schema: No object selected

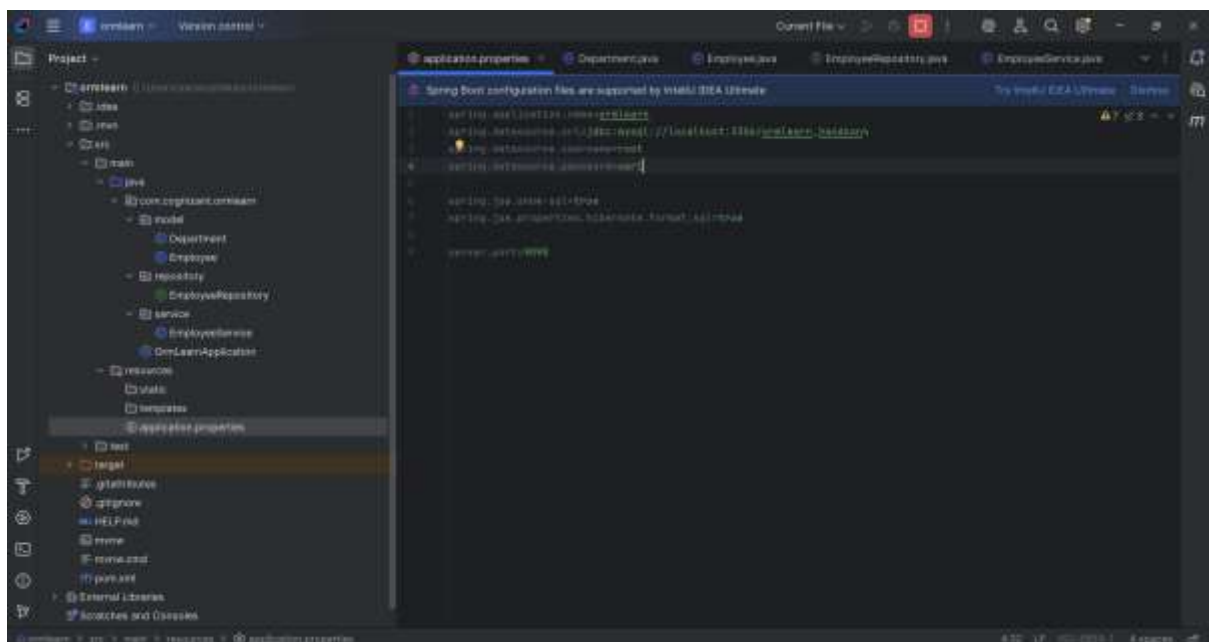
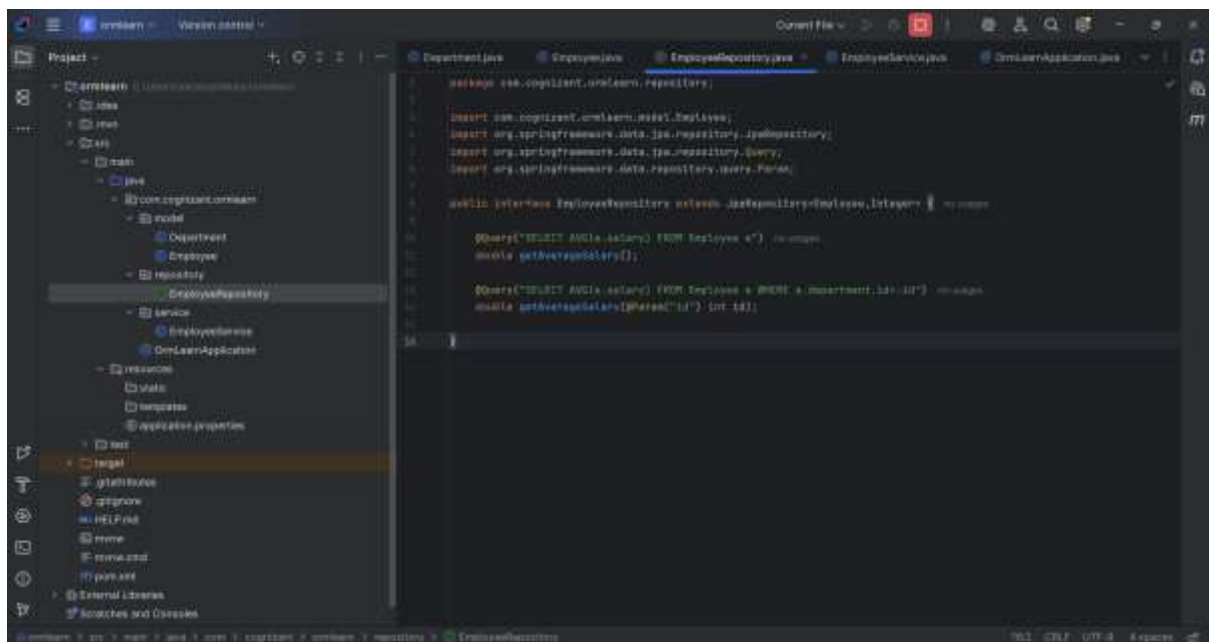
```

177 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 1, 0.5); -- sq_id=1
178 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 2, 1.0); -- sq_id=2
179 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 3, 0.5); -- sq_id=3
180 * INSERT INTO attempt_question (at_id, qd_id, sq_score) VALUES (1, 4, 0.5); -- sq_id=4
181
182 -- Q1 for class (sq_id=1) - wrong answer
183 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 1, 1); -- 1 not wrong
184 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 2, 0); -- 0 not selected
185 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 3, 0); -- 0 not selected
186 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (1, 4, 0); -- 0 not selected
187
188 -- Q2 for class (sq_id=2) - correct
189 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 1, 0); -- 0 not selected
190 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 2, 0); -- 0 not selected
191 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 3, 1); -- 1 total selected (correct)
192 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (2, 4, 0); -- 0 not selected
193
194 -- Q3 for class (sq_id=3) - wrong
195 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 1, 0); -- 0 not selected
196 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 2, 1); -- 1 total selected (wrong)
197 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 3, 0); -- 0 not selected
198 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (3, 4, 0); -- 0 not selected
199
200 -- Q4 for class (sq_id=4) - partial
201 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (4, 1, 0); -- 0 not selected
202 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (4, 2, 0); -- 0 not selected
203 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (4, 3, 1); -- 1 total selected (partial)
204 * INSERT INTO attempt_option (aq_id, op_id, sq_answer) VALUES (4, 4, 0); -- 0 not selected
205

```





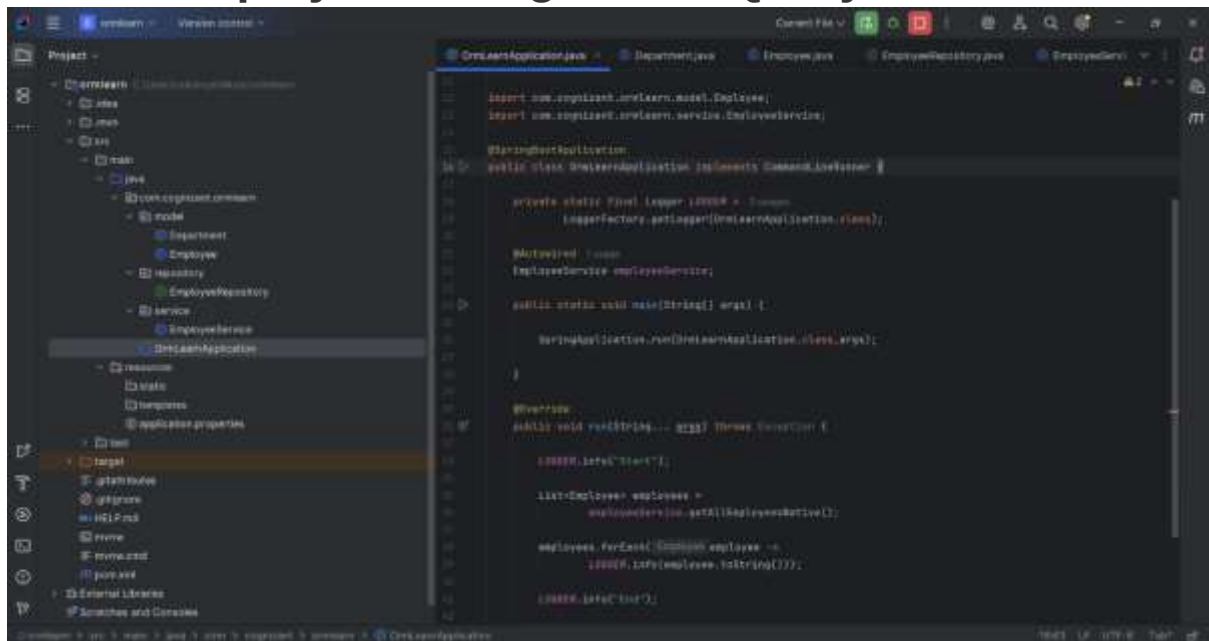
OUTPUT:

```
2024-07-23T09:13:36.617+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Start
select
  avg(e1.S.sal_salary)
from
  employee e1_0
2024-07-23T09:13:36.927+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Average Salary : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.966+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.973+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.981+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
2024-07-23T09:13:36.981+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | End
```

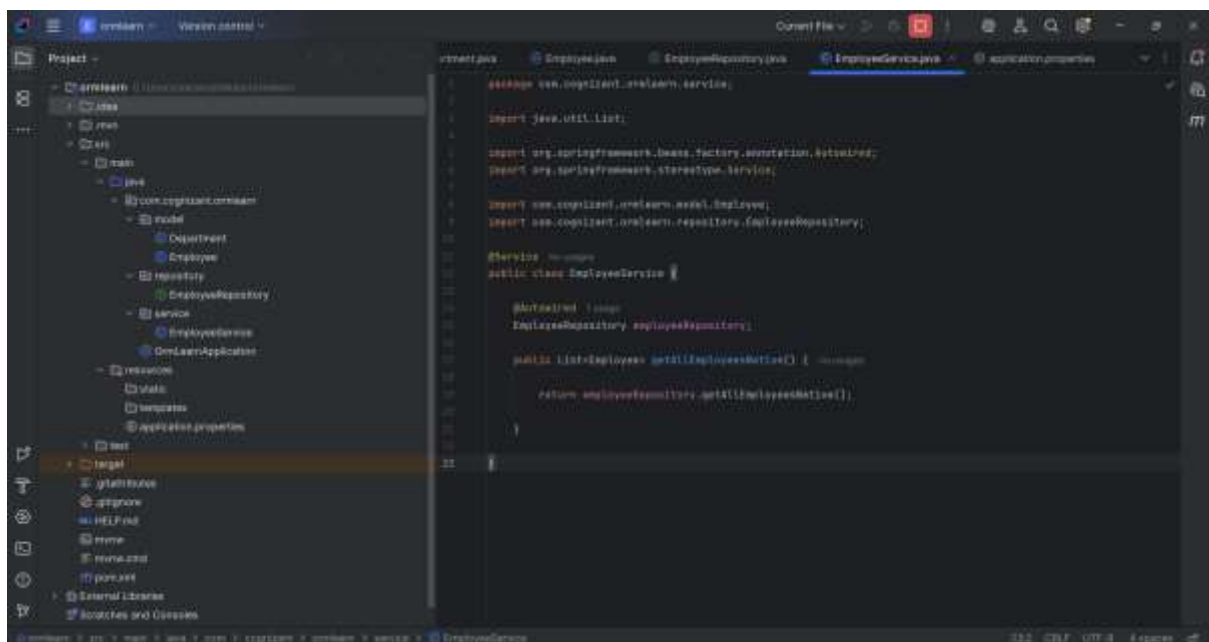
```
2024-07-23T09:13:36.617+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Start
select
  avg(e1.S.sal_salary)
from
  employee e1_0
2024-07-23T09:13:36.927+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Average Salary : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.966+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.973+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
select
  avg(e1.S.sal_salary)
from
  employee e1_0
where
  e1.D.dept=?
2024-07-23T09:13:36.981+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | Department Average : 33333.3
2024-07-23T09:13:36.981+02:00 INFO 13488 --- [ormlearn] | main | c.s.ormlearn.OrmLearnApplication | End
```

Hands on 5

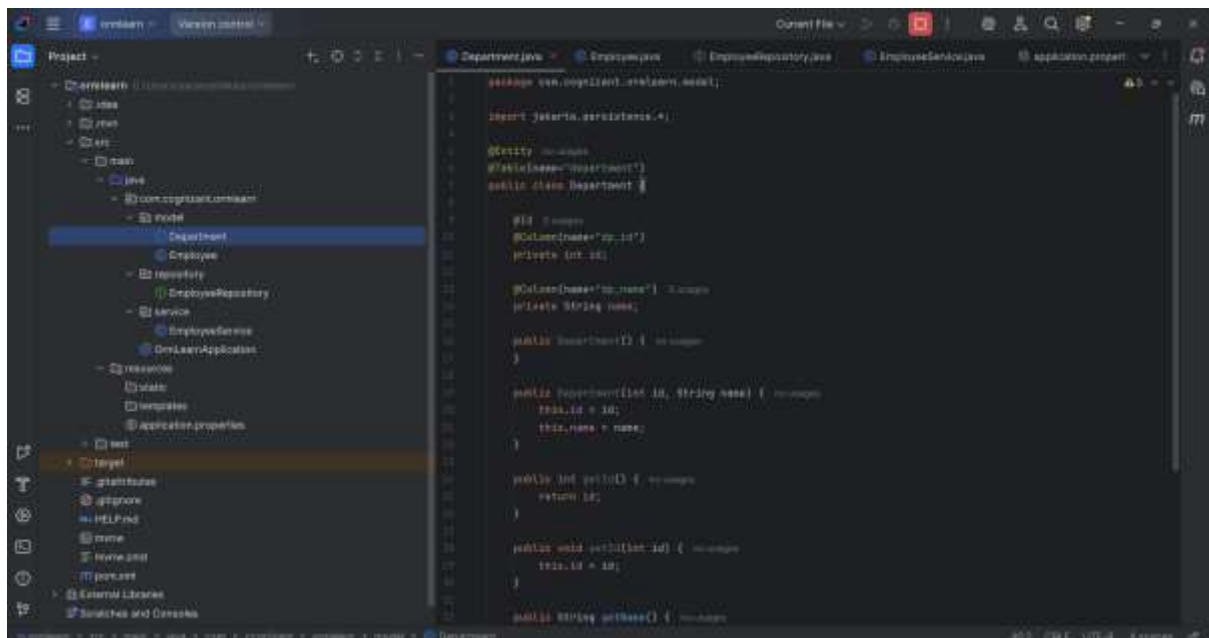
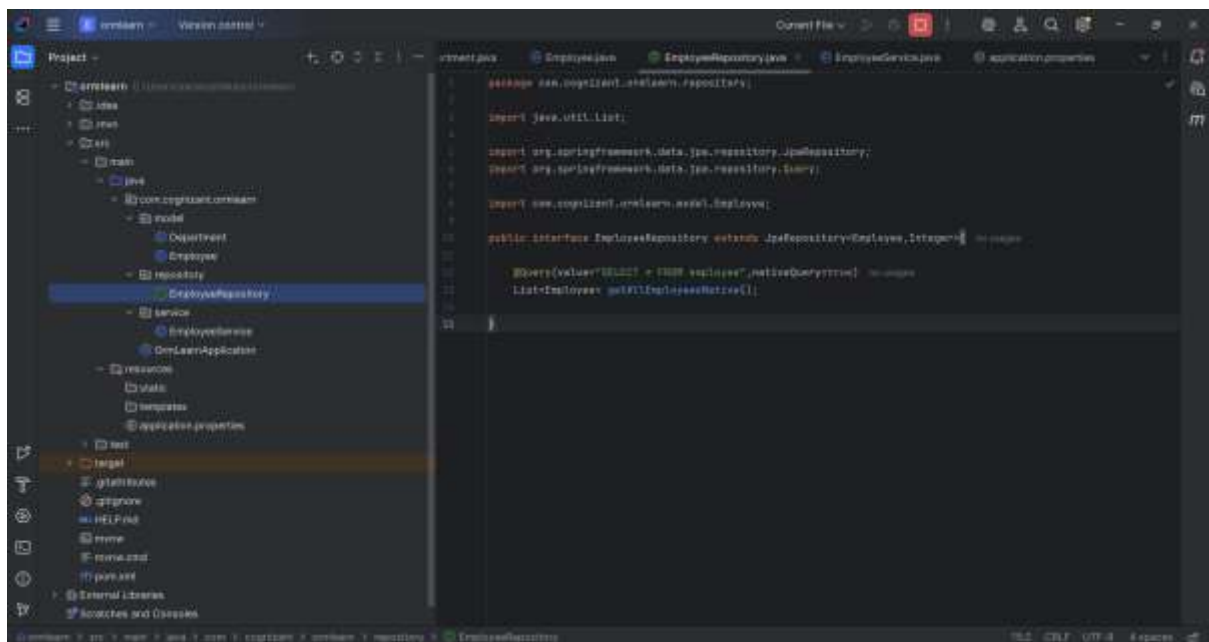
Get all employees using Native Query

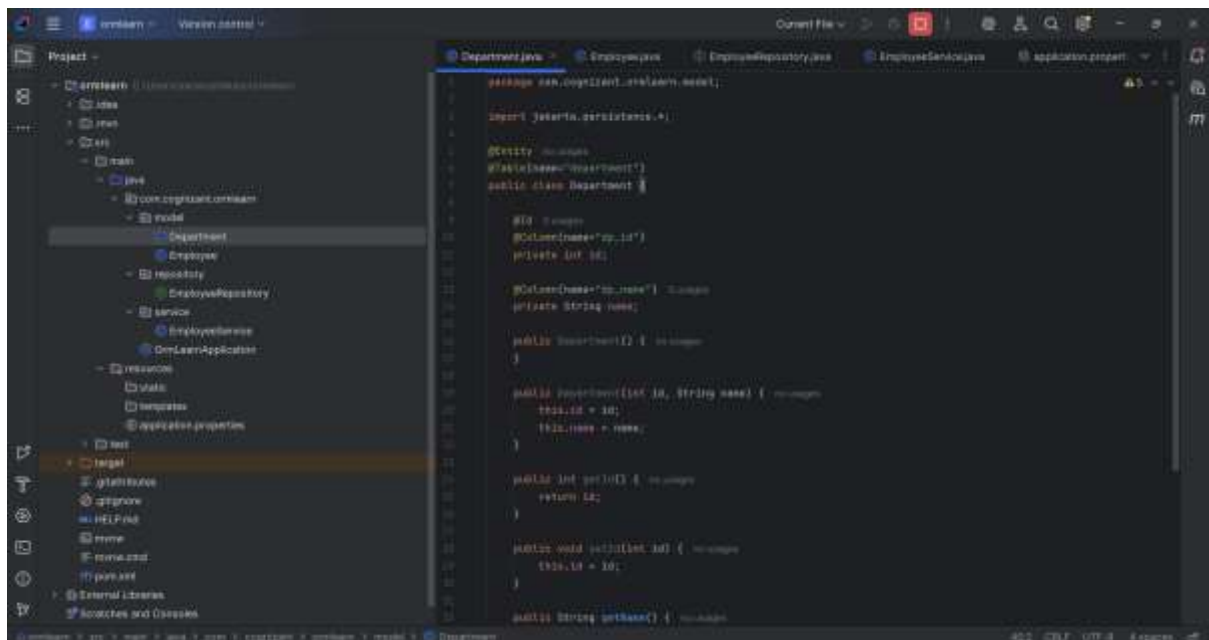
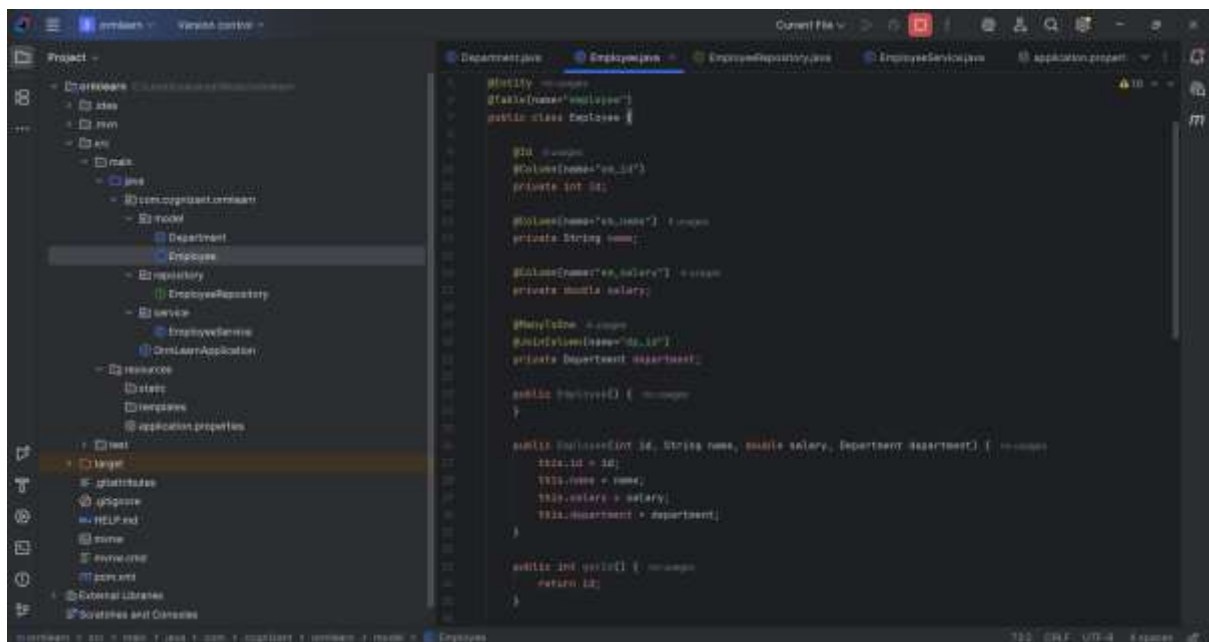


```
1 import com.cognizant.emilearn.model.Employee;
2 import com.cognizant.emilearn.service.EmployeeService;
3
4 @SpringBootTest
5 public class EmilearnApplication implements CommandLineRunner {
6
7     private static final Logger LOGGER = LogManager.getLogger(EmilearnApplication.class);
8
9     @Autowired
10     EmployeeService employeeService;
11
12     public static void main(String[] args) {
13         SpringApplication.run(EmilearnApplication.class, args);
14     }
15
16     @Override
17     public void run(String[] args) throws Exception {
18         LOGGER.info("Start");
19
20         List<Employee> employees =
21             employeeService.getAllEmployees();
22
23         employees.forEach(employee ->
24             LOGGER.info(employee.toString()));
25
26         LOGGER.info("End");
27     }
28 }
```

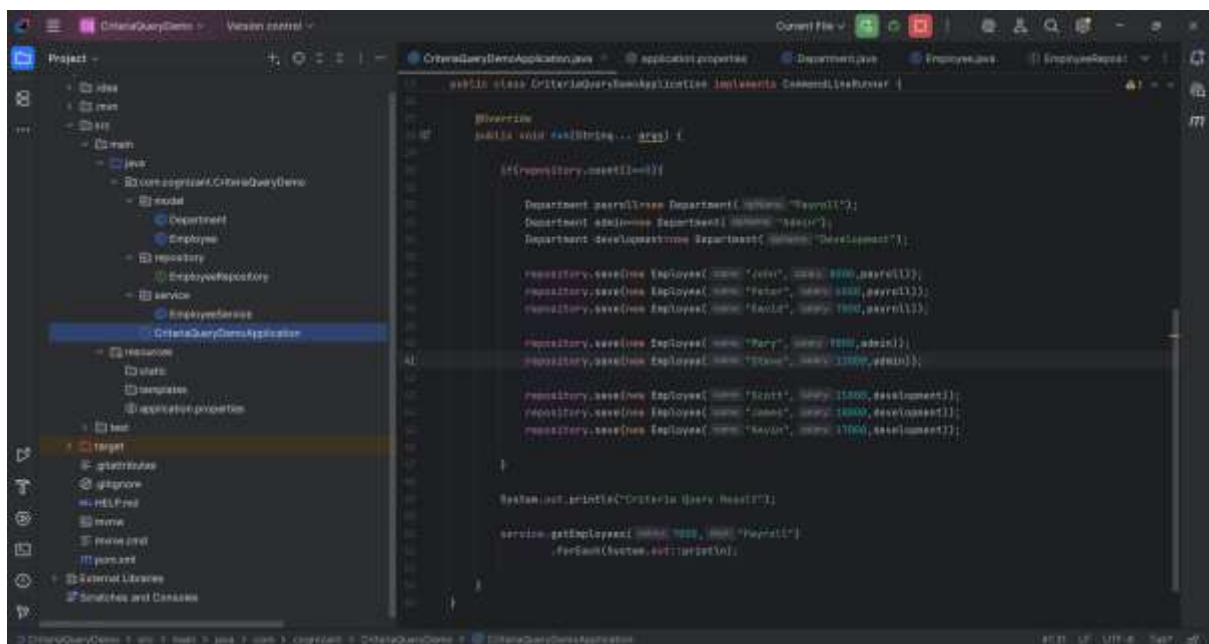
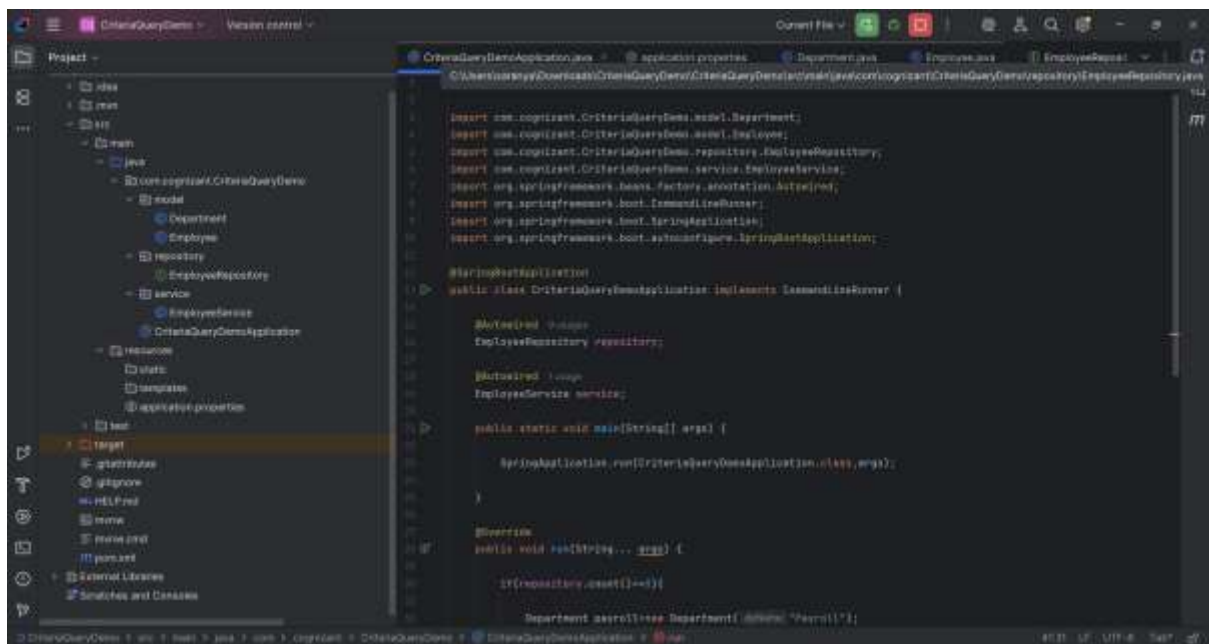


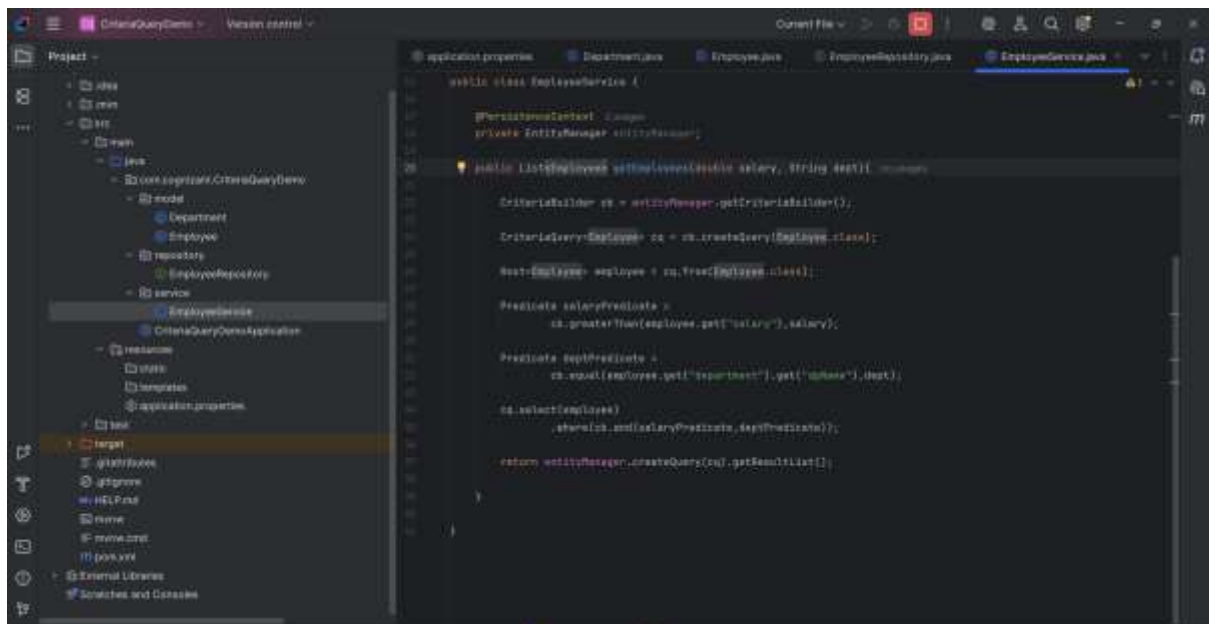
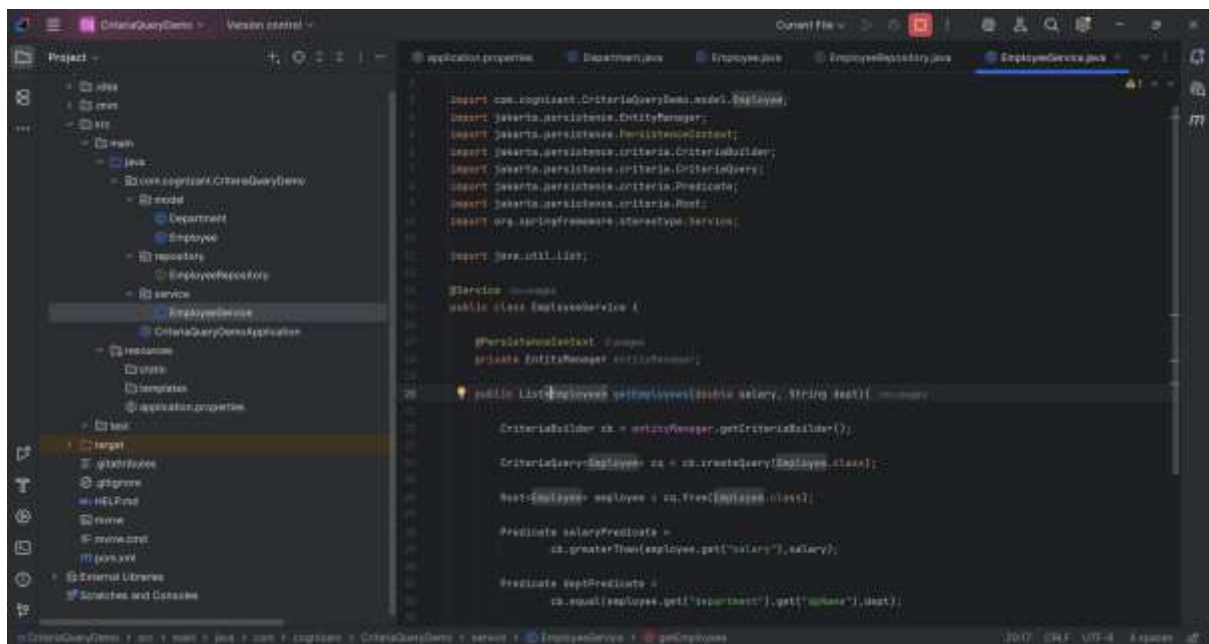
```
1 package com.cognizant.emilearn.service;
2
3 import java.util.List;
4
5 import org.springframework.data.repository.query.support;
6 import org.springframework.stereotype.Service;
7
8 import com.cognizant.emilearn.model.Employee;
9 import com.cognizant.emilearn.repository.EmployeeRepository;
10
11 @Service
12 public class EmployeeService {
13
14     @Autowired
15     EmployeeRepository employeeRepository;
16
17     public List<Employee> getAllEmployees() {
18         return employeeRepository.findAll();
19     }
20 }
```

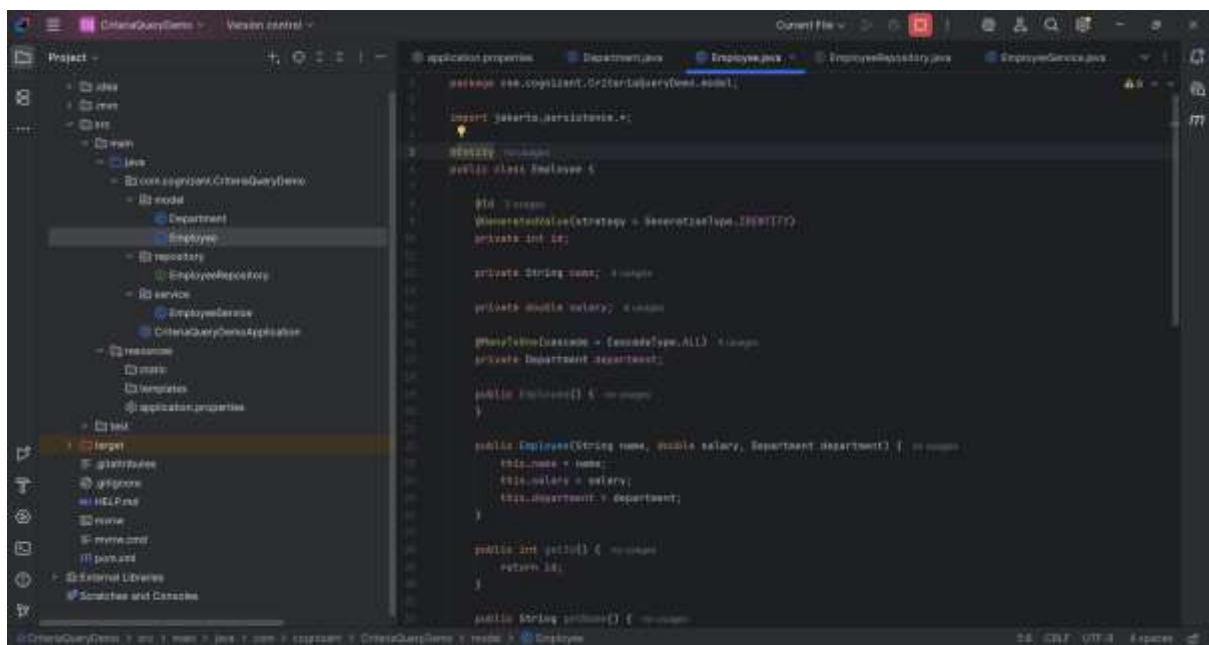
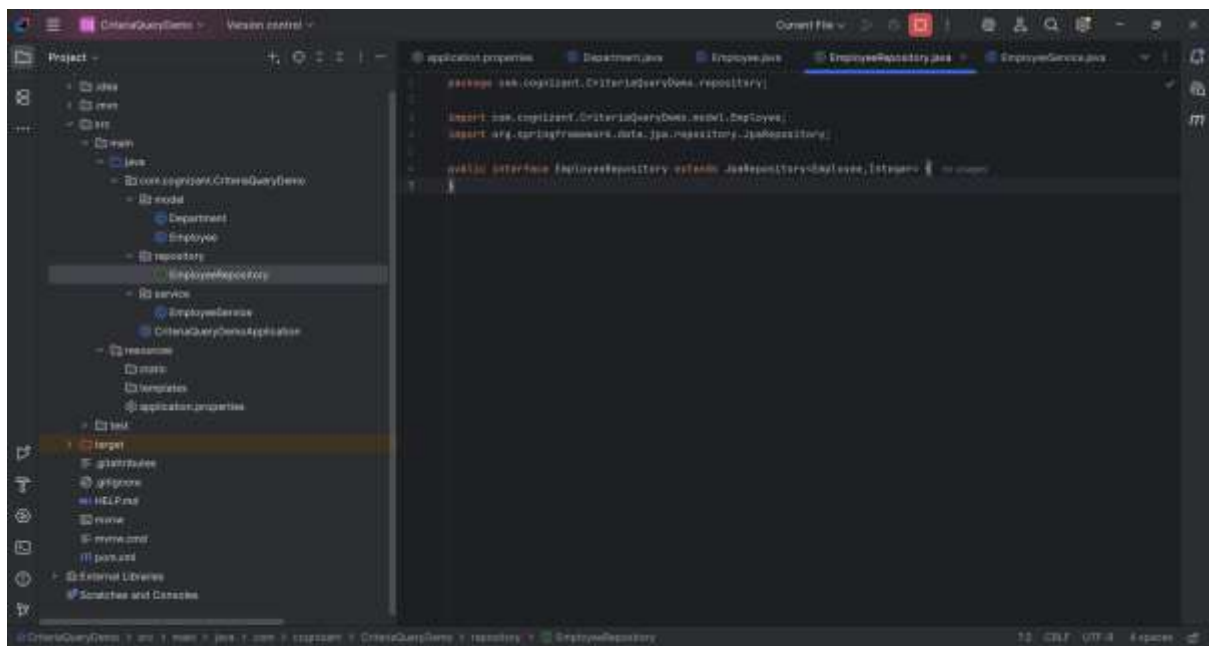


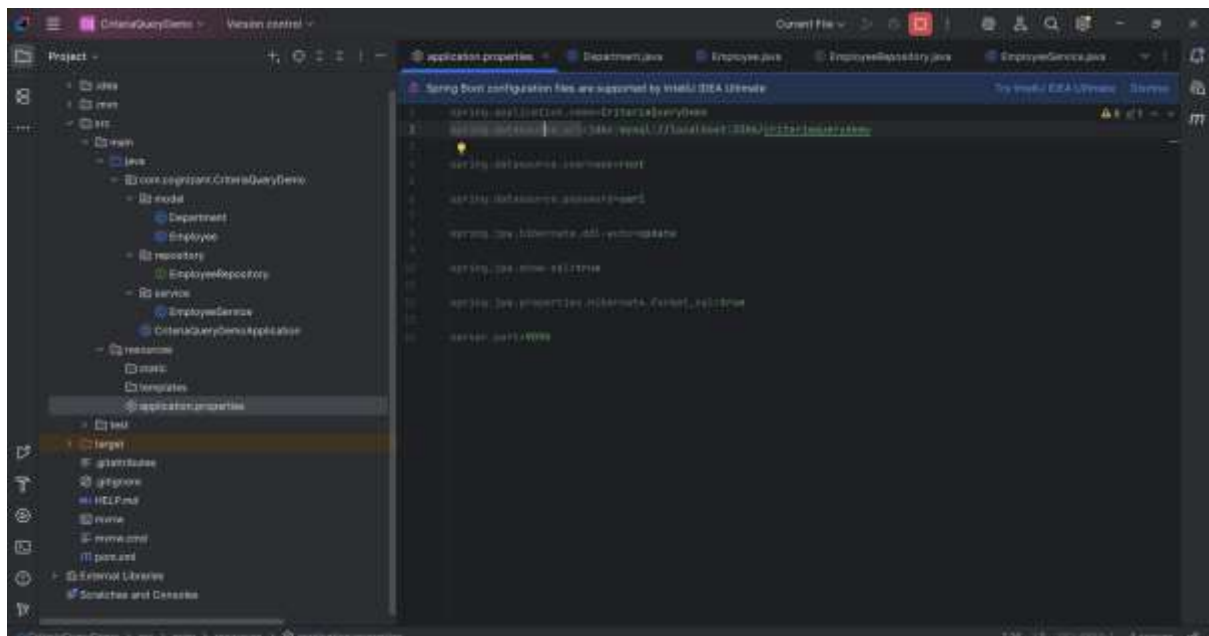
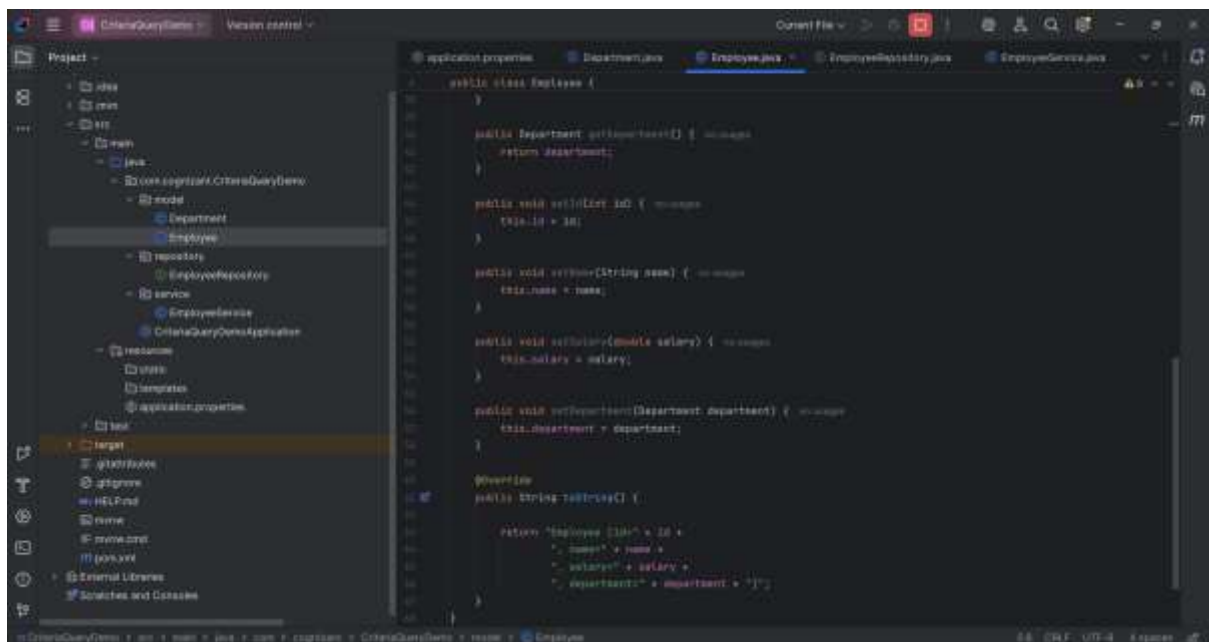


Criteria Query









OUTPUT:

